

WEEK
6

Oaks and Oak-Northern Hardwood Transition

Instruction | retrieval | multi-organ practice | field-ready reference

Public learner edition

This packet contains the teaching guide, cumulative reference, and formative practice for the week. Secure graded assessments remain in the online course so unfamiliar examination specimens and answer material are not published.

Student / participant

Date started

Planned pace / study notes

Field standard

Identify the narrowest rank the visible evidence can defend. A common name OR a scientific name satisfies an ordinary identity prompt unless the activity explicitly tests nomenclature. State uncertainty instead of converting weak evidence into a confident species claim.

Use this packet alongside the online course at cochetopa.co/course/. Work the instruction first, complete retrieval without looking back, and use the secure online assessment when ready for grading.

Week 6 overview

Learning objectives

1. Discriminate the regional red-oak and white-oak groups from leaves, buds, bark, acorns, and site.
2. Identify oak specimens across sun and shade foliage, age, and regional context.
3. Use acorn cup, bud, lower-leaf, and bark characters instead of relying on lobe shape alone.
4. Connect mast, establishment, fire, succession, and silvicultural response to field management.
5. Maintain cumulative northern-hardwood and conifer retrieval while oak density increases.

Species introduced this week

Common name	Scientific name	Family	Tier
Northern red oak	Quercus rubra	Fagaceae	A
Black oak	Quercus velutina	Fagaceae	B
Scarlet oak	Quercus coccinea	Fagaceae	B
Northern pin oak	Quercus ellipsoidalis	Fagaceae	B
Pin oak	Quercus palustris	Fagaceae	B
White oak	Quercus alba	Fagaceae	A
Swamp white oak	Quercus bicolor	Fagaceae	B
Bur oak	Quercus macrocarpa	Fagaceae	B
Chestnut oak	Quercus montana	Fagaceae	B
Common hackberry	Celtis occidentalis	Cannabaceae	B

Cumulative review

63 taxa remain eligible. Earlier species do not disappear from retrieval merely because the weekly theme changes.

Confusion sets

Sugar / black / Norway maple: Petiole latex; leaf underside hair and stipules; bud size/shape; samara angle

Request: Fresh-broken petiole; lower surface; terminal bud; ripe samara

Red / silver maple: Sinus depth; lower-surface whiteness; spring samara size; twig odor; bark strips

Request: Complete leaf underside; samara with scale; crushed twig; mature bark

Striped / mountain maple: Green-white striped bark versus unstriped; leaf size/base; fruit cluster posture; growth form

Request: Young stem; intact leaf; fruit cluster; whole plant

White / green / black ash: Leaf-scar shape and bud position; leaflet stalks; samara wing extent; black-ash corky bark

Request: Twig node and leaf scar; whole rachis; samara; bark; hydrologic setting

Paper / heart-leaved / gray birch: Leaf base and vein count; twig resin glands; peeling versus tight bark; black branch marks; elevation

Request: Several mature leaves; resin-warty twig; bark sheet; whole growth form; location

Yellow / sweet birch: Both wintergreen; yellow birch golden curling bark and spur shoots versus sweet birch dark blocky bark

Request: Scratched twig; young and mature bark; fruiting strobile; lower leaf surface

White / black / red / Norway spruce: Cone size and scale margin; twig hairs; bud resin; foliage color/odor; pendulous branchlets

Request: Full cone; twig macro; bud; whole crown; regional site

Spruce / fir / hemlock: Woody pegs versus pad base versus petiole; needle cross-section; cone orientation and persistence

Request: Needle attachment macro; underside; terminal shoot; intact or attached cone

Red / jack / Scots / pitch pine: Fascicle count/length; bend test; cone curvature/prickles; upper-bole orange bark; epicormic shoots

Request: Several complete fascicles; cone and umbo; upper and lower bark; whole crown

Tamarack / European larch: Cone size and scale count/edge; needle cluster density; native peatland versus planting

Request: Full cone; spur shoot; cluster count; site/planting context

Red-oak group: Bud hair/size; lower-leaf hair; sinus architecture; acorn cup depth/rings; bark; site and branch angle

Request: Several sun leaves; terminal buds; complete acorn/cup; mature bark; whole crown

White-oak group: Lower-leaf felt; fruit stalk; fringed cup; chestnut teeth; bark architecture; site

Request: Lower surface; acorn/cup/stalk; terminal bud; upper and lower bole; site

Aspens / balsam poplar / cottonwood: Petiole flattening; tooth size; leaf base; bud size/stickiness/odor; bark and floodplain/upland context

Request: Complete leaf and petiole; bud; young and old bark; clonal form; site

American / slippery / rock / Siberian elm: Leaf roughness/base; bud hair; corky wings; samara hair/notch; bark layering; crown

Request: Leaf surfaces; buds; samara; corky twigs; existing bark section; crown

Black / pin / chokecherry: Fruit cluster architecture; petiole glands; lower-midvein hair; bark age; whole pioneer/clonal form

Request: Raceme or cluster; petiole macro; lower leaf; mature bark; whole plant

American hornbeam / eastern hophornbeam: Muscular smooth bole and three-lobed fruit bracts versus shreddy bark and hoplike sacs

Request: Bole; fruit; bud; wet versus dry site

Yellow birch / black cherry bark: Golden horizontal curls and wintergreen twig versus burnt-chip plates, petiole glands, and almond twig odor

Request: Young and old bark; scratched twig; leaf petiole and underside; fruit

Serviceberry / mountain-ashes / chokecherry: Simple versus compound leaves; pome calyx versus drupe; leaflet proportions; buds; fruit architecture

Request: Whole leaf; fruit close-up; bud; flowers; multiple leaves

Assessment schedule

Assessment	Type	Items	Time
W06-PRACTICAL	weekly identification practical	24	48 min
W06-NOMENCLATURE-3	cumulative nomenclature check	40	30 min
W06-SILVICS-3	cumulative silvics retrieval	20	50 min

Five-session field workflow

Session A - Regional red-oak group

character-combination teaching across sun and shade foliage. Modalities: foliage, bark, reproductive, dormant, whole tree context.

☐ Session A completed (enter X)

Session B - Red-oak retrieval

unfamiliar isolated organs and whole-tree decisions. Modalities: foliage, bark, reproductive, dormant, whole tree context.

☐ Session B completed (enter X)

Session C - White-oak group and hackberry

direct group contrasts followed by mixed isolated specimens. Modalities: foliage, bark, reproductive, dormant, whole tree context.

☐ Session C completed (enter X)

Session D - Oak multi-organ laboratory

leaves buds acorns bark and site stations. Modalities: foliage, bark, reproductive, dormant, whole tree context.

☐ Session D completed (enter X)

Session E - Oak-transition cumulative retrieval

adaptive retrieval with nomenclature and silvics prompts. Modalities: foliage, bark, reproductive, dormant, whole tree context.

☐ Session E completed (enter X)

Evidence-to-decision protocol

Step	Field action
1. Inventory	Name only what is visible: organ, arrangement, attachment, surface, scale, maturity, damage, and context.
2. Generate	Name the leading identity and the nearest plausible alternative before seeking confirmation.
3. Separate	Cite the visible character that changes the decision; do not substitute habitat or color alone for structure.
4. Calibrate	Use species, genus, or insufficient evidence according to the photograph, not according to what you hope it shows.
5. Request	Ask for the single additional view with the highest expected diagnostic value.

Part I - Species instruction atlas

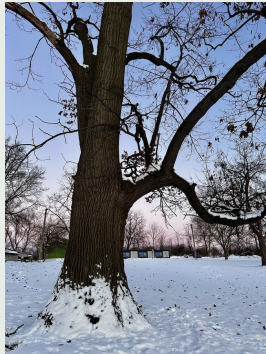
These profiles are labeled teaching material. Photographs come only from the teaching pool; no exact image in the formal assessment section appears here.

Northern red oak - *Quercus rubra*

New this week | Family: Fagaceae | Tier A | Priority 1 - core

Widespread from MN/WI/MI and southern ON/QC through NY/New England/NB and PA/NJ

Major northern oak and valuable timber/mast tree; responds well to release



Bark

(c) Ben Redding, some rights reserved...
INAT_PHOTO_597845529 | cc-by



Foliage

no rights reserved
INAT_PHOTO_713498929 | cc0



Mixed Evidence

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Evidence	Diagnostic interpretation
Foliage	Alternate leaves usually have 7-11 bristle-tipped lobes, sinuses generally extending less than halfway to the midrib, and a broad wedge-shaped to shallowly heart-shaped base. The upper surface is dull to moderately glossy green; the paler lower surface is mostly hairless except for possible small axillary tufts. Confirm the bristles because damaged or late-season tips can be obscure.
Bark age series	Young stems are smooth gray-brown. Saplings and pole timber develop long smooth pale vertical bands between shallow fissures; on mature boles these become broad, flat-topped ridges with shiny stripe-like tracks often called ski trails. Very old bark darkens and breaks into wider ridges, so the striped character may be confined to upper bole sections.
Dormant	Stout reddish-brown to gray-brown twigs end in a cluster of several pointed, reddish-brown, mostly hairless buds. Leaf scars are alternate and semicircular with numerous bundle scars. The clustered terminal buds establish oak, while relatively smooth buds and twigs favor northern red oak over black oak but do not alone exclude scarlet or northern pin oak.
Fruit / cone / seed	Red-oak-group acorns require two growing seasons. The broad ovoid nut is commonly 19-30 mm long and sits in a shallow, flat saucer cup that usually covers one-quarter or less of the nut; the tightly appressed cup scales lack a conspicuous fringe. A nearly mature first-year green acorn must not be confused with a one-season white-oak-group fruit.
Whole tree / context	A large, straight-boled canopy tree with a rounded crown, common on well-drained middle and lower slopes, coves, benches, and deep loams. It also occupies drier uplands, but good height and form on mesic northern-hardwood sites are strong contextual support across most of the course region.
Variation / condition	Sun leaves may be smaller and more deeply cut than shade leaves, epicormic and stump-sprout leaves may be oversized, and acorn form varies geographically. Northern red oak hybridizes or introgresses with other red-oak-group species. Oak wilt, bacterial leaf scorch, drought, defoliators, and late-season senescence can all scorch or shed foliage; those symptoms describe condition, not identity.
Evidence limit	The best species call combines several mature leaves, a terminal-bud cluster, mature acorn and cup, bark age, and range/site. Ski-trail bark can strongly support a field call on a typical mature bole, but one bark panel, one variable leaf, or an immature acorn cannot reliably separate the red-oak-group complex. Record intermediates as <i>Quercus</i> section <i>Lobatae</i> or a named species complex when the decisive cup, bud, and lower-leaf views are absent.

Confuser control

Alternative	Visible separator	Resolving view
Quercus velutina	Black oak more often has a densely hairy, angular terminal bud, yellow to orange inner bark, lower-leaf pubescence or rusty axillary tufts, and a deeper cup with loose or fringed upper scales. Northern red oak usually has smoother buds, paler nearly hairless lower leaves, ski-trail bark, and a shallow saucer cup.	Terminal buds in scale, lower leaf surfaces, a mature cup in side view, intact mature bark and, only if sampling is authorized, a tiny view of freshly exposed inner bark.
Quercus coccinea	Scarlet oak typically has deeper rounded or C-shaped sinuses and a cup enclosing about one-third to one-half of the nut, often with concentric rings around the nut tip. Northern red oak usually has shallower sinuses and a broad shallow cup; leaf depth alone overlaps.	Several crown leaves, a complete mature acorn showing cup depth and nut tip, terminal buds, bark above breast height, and slope position.
Quercus ellipsoidalis	Northern pin oak is centered on dry nutrient-poor sands of the western Great Lakes and usually has very deep sinuses plus a deeper cup around a commonly ellipsoid nut. Northern red oak ranges much more widely, favors better mesic sites, and usually has the shallow saucer cup; hybrids make single organs unsafe.	Mature acorns from the same tree, multiple leaves, terminal buds, georeferenced range, soil or landform context, and neighboring oaks.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Intermediate in shade tolerance; seedlings can persist for a time beneath partial canopy but durable recruitment and crown development require release.
Moisture / soils	Best on deep, fertile, moist but well-drained loams and silt loams; it occupies a broad site range but performs poorly on prolonged saturation and the most droughty sterile sands.
Geographic affinity	The broadest and most northerly major eastern red oak, extending throughout the course region and into the Acadian forest fringe.
Associations	Occurs with sugar maple, basswood, white ash, beech, yellow birch, hemlock, white pine, black cherry, white oak, black oak, and hickories according to site and region.
Regeneration	Animal-cached acorns and abundant stump sprouts contribute; successful stands commonly originate from established advance reproduction or sprouts released by substantial canopy disturbance.
Vegetative reproduction	Young and intermediate stems sprout strongly from the stump or root collar after cutting or top-kill; sprouting capacity declines with increasing size and age.
Seed / mast	Acorns mature in their second season, with irregular crops and episodic heavy mast; insects, deer, rodents, birds, and fungi remove a large share before establishment.
Establishment	A mineral-soil or mixed-litter seedbed, adequate moisture, acorn protection, and enough light after establishment are important; small seedlings rarely compete through dense fern, shrub, or shade-tolerant advance layers.
Succession	Mid-successional to persistent canopy species maintained by disturbance; without adequate disturbance it is often replaced in regeneration layers by sugar maple, beech, and other shade-tolerant trees.
Disturbance response	Codominant young trees respond well to timely crown release. Fire top-kills small stems but can favor oak through sprouting and competitor reduction; severe repeated fire, wounding, or soil compaction is damaging.
Longevity / growth	Moderate to fast for an oak on good sites, commonly living 200-300 years and occasionally longer, with strong height growth after adequate release.
Browse	Acorns, seedlings, and sprouts are heavily used by deer and other wildlife; repeated browse can prevent otherwise abundant advance reproduction from reaching recruitment height.
Insects / disease	Oak wilt is a major concern in susceptible red-oak-group stands, while gypsy moth or spongy moth defoliation, two-lined chestnut borer, acorn weevils, shoestring root rot, and leaf diseases can be important. Symptoms require diagnosis independent of species identification.
Management	Inventory advance regeneration by size, not seedling count alone; retain productive seed trees and vigorous crop trees, protect roots and boles, manage browse and competing vegetation, and avoid forcing hybrid-looking stems into a species label.
Silvicultural systems	Shelterwood and group or even-aged methods that build large advance reproduction and then provide substantial light are generally more reliable than repeatedly light selection cuts; early crop-tree release can increase diameter growth, whereas late heavy release may trigger epicormic branches.

Personal field cue for Northern red oak

learner-created retrieval hook

Black oak - *Quercus velutina*

New this week | Family: Fagaceae | Tier B | Priority 2 - regional

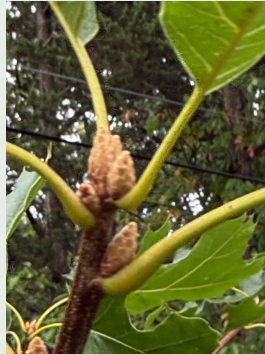
Southern Great Lakes, southern ON, NY, PA/NJ, and southern New England

Dry-site red-oak-group associate and timber tree



Bark

(c) Skyler Principe, some rights...
INAT_PHOTO_473539413 | cc-by



Dormant

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INAT_PHOTO_719075749 | cc0



Foliage

(c) Laurel Meredyk, some rights...
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Evidence	Diagnostic interpretation
Foliage	Alternate leaves are extremely variable, usually with 5-9 bristle-tipped lobes and sinuses ranging from shallow to deep. The upper surface is dark glossy green; the lower surface is commonly yellow-green to tawny with diffuse pubescence or conspicuous rusty tufts in vein axils. Use upper-crown mature leaves because vigorous sprouts can be atypical.
Bark age series	Young bark is smooth gray. It darkens early and becomes thick, deeply furrowed, and blocky on mature stems, often lacking the long smooth ski trails typical of northern red oak. Freshly exposed inner bark is yellow to orange because of quercitron pigment, but cutting bark is destructive and should be used only on authorized samples.
Dormant	Twigs are moderately stout and reddish-brown to gray, ending in a cluster of relatively large, angular or ovoid terminal buds densely covered with gray-tawny hairs. Bud hair is one of the best non-destructive winter contrasts with northern red oak, although occasional buds are less densely hairy.
Fruit / cone / seed	Acorns mature in two seasons. The ovoid nut is about 13-20 mm long and is enclosed one-third to one-half by a deep bowl- to top-shaped cup; lower scales are appressed but scales near the rim are often loose, ragged, or fringed. Cup depth and scale posture are more useful than nut color.
Whole tree / context	A medium to large canopy oak with an open irregular crown, frequent on dry upper slopes, ridges, sandy or rocky uplands, and glacial soils in the southern part of the course region. It can grow well on rich sites, so a dry ridge supports but does not establish identity.
Variation / condition	Leaf depth and lower-surface hair vary greatly, and black oak hybridizes with northern red, scarlet, and northern pin oaks where ranges meet. Bark color changes with moisture and lichen; inner-bark yellow fades after exposure. Oak wilt and defoliation can alter the crown but cannot distinguish black oak from other susceptible red oaks.
Evidence limit	A defensible call combines hairy terminal buds, lower-leaf pubescence, deep loose-scaled cup, blocky mature bark, and regional context. Yellow inner bark is supporting evidence only when legally and ethically exposed and is not required. A variable leaf or dark blocky bark alone should remain red-oak group.

Confuser control

Alternative	Visible separator	Resolving view
<i>Quercus rubra</i>	Northern red oak usually has mostly hairless pointed buds, a shallow saucer cup, paler nearly hairless leaf undersides, and mature bark with long smooth vertical tracks. Black oak tends to hairy buds, a cup enclosing one-third to one-half of the nut, lower-leaf hair, and yellow-orange inner bark.	Terminal buds, lower leaf surfaces, mature cup in profile, intact bole bark and, if an authorized wound already exists, the fresh inner-bark color.
<i>Quercus coccinea</i>	Scarlet oak more often has deeply cut leaves with clean pale undersides except small axillary tufts and a tight-scaled cup with concentric rings around the nut tip. Black oak more often has broader variable lobes, more diffuse lower-surface hair, densely hairy buds, and loose scales at the cup rim.	Several upper-crown leaves, terminal buds, lower leaf surfaces, and complete mature acorns showing the cup rim and nut tip.
<i>Quercus ellipsoidalis</i>	Northern pin oak is concentrated on nutrient-poor western Great Lakes sands, has commonly ellipsoid nuts and often more deeply cut, nearly hairless leaves. Black oak extends farther south and east and more often shows hairy buds, lower-leaf pubescence, and yellow inner bark; introgression can erase the boundary.	Georeferenced range and soil context, terminal buds, lower leaf surfaces, a series of mature acorns, and neighboring oak identities.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Intermediate but light-demanding for successful recruitment; seedlings may persist under partial cover but do not maintain growth beneath dense shade.
Moisture / soils	Occurs from dry sandy and rocky uplands to moist rich well-drained loams; it avoids prolonged saturation and is often most competitive on moderately dry acidic sites.
Geographic affinity	Eastern and central species reaching its northern limits in southern Ontario, the southern Great Lakes, New York, and New England.
Associations	Commonly associates with northern red, scarlet, white, and chestnut oaks, hickories, blackgum, red maple, white pine, and dry-site understory shrubs.
Regeneration	Animal-dispersed acorns create seedlings and advance reproduction; substantial overstory opening and control of competing shade-tolerant vegetation are usually needed for canopy recruitment.
Vegetative reproduction	Stumps of small and medium stems sprout vigorously after cutting or top-kill, often contributing more reliable regeneration than seedlings on disturbed sites.
Seed / mast	Second-season acorns are produced irregularly, with good crops separated by lighter years; acorn insects and wildlife consume much of the crop.
Establishment	Acorns need contact with moist mineral soil or thin litter and protection from desiccation and predators; established seedlings need increasing light before competitors overtop them.
Succession	A disturbance-dependent, mid-successional canopy species that can persist on dry edaphic sites but commonly loses regeneration space to more shade-tolerant hardwoods during fire exclusion.
Disturbance response	Young trees and seedlings sprout after fire or cutting; thicker bark improves survival with size. Moderate disturbance can favor it, while repeated severe fire, wounding, drought plus defoliation, or compaction can cause decline.
Longevity / growth	Moderate growth and commonly less than 200 years on poor sites, becoming a large valuable tree on favorable soils but generally shorter-lived than white oak.
Browse	Acorns and young shoots are important wildlife foods, and deer browsing can suppress advance regeneration and sprout height growth.
Insects / disease	Oak wilt can rapidly kill red-oak-group trees; spongy moth defoliation, two-lined chestnut borer, shoestring root rot, cankers, and acorn insects are also important, particularly when drought-stressed.
Management	Use site and multi-organ evidence before marking species, protect sound seed and crop trees, retain adequate advance regeneration or sprout potential, and reduce browse and understory competition before final overstory removal.
Silvicultural systems	Shelterwood, group openings, or other even-aged treatments can recruit black oak when they deliberately establish and release reproduction; coppice contributions are strong on younger stands, whereas low-intensity selection tends to favor tolerant competitors.

Personal field cue for Black oak

learner-created retrieval hook

Scarlet oak - *Quercus coccinea*

New this week | Family: Fagaceae | Tier B | Priority 2 - regional

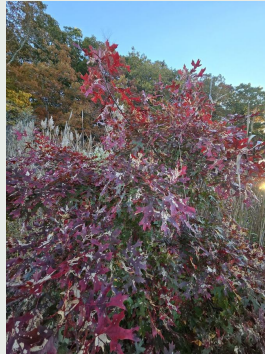
PA/NJ, southern NY, and southern New England; southern Great Lakes fringe

Shade-intolerant fire-associated oak of dry acidic ridges and barrens



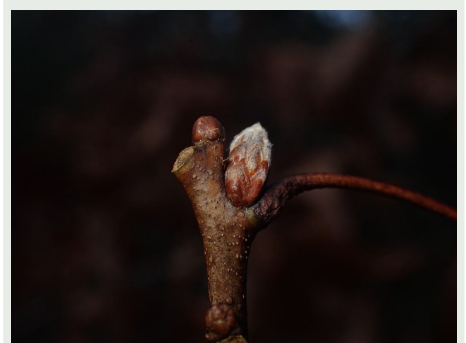
Bark

(c) John Baur, some rights reserved (CC...
INAT_PHOTO_342701099 | cc-by



Condition

no rights reserved
INAT_PHOTO_585005166 | cc0



Dormant

(c) John Baur, some rights reserved (CC...
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Evidence	Diagnostic interpretation
Foliage	Alternate glossy leaves usually have 7-9 narrow bristle-tipped lobes and deep broad C- or U-shaped sinuses extending well over halfway to the midrib. The lower surface is pale and mostly hairless apart from small axillary tufts. Fall color can be vivid scarlet but is too variable and seasonal to identify the species.
Bark age series	Sapling bark is smooth gray-brown. Mature bark becomes dark gray to nearly black with irregular narrow ridges and shallow to moderate fissures; the red to pinkish inner bark lacks black oak's strong yellow pigment. Exterior bark overlaps black and northern red oak and is weakest on young or rain-darkened stems.
Dormant	Slender reddish-brown twigs end in clustered ovoid, pointed reddish-brown buds that may be finely pale-hairy toward the tip but are not densely woolly like typical black oak. Persistent dead lower branches may occur but are shared with northern pin and pin oaks.
Fruit / cone / seed	Acorns mature in two seasons. The 13-25 mm ovoid nut sits one-third to one-half deep in a bowl-shaped cup with tightly appressed scales; concentric rings around the exposed nut tip are a useful scarlet-oak character when clearly visible. Confirm a naturally attached or confidently associated fruit because ground acorns mix beneath oak crowns.
Whole tree / context	A medium to large light-demanding oak with an open irregular crown, characteristic of dry acidic upper slopes, ridges, barrens, and fire-influenced oak-pine woods in Pennsylvania, New Jersey, southern New York, and southern New England. A basal swell and retained dead branches may occur but are not unique.
Variation / condition	Shade leaves may be broader and less deeply cut, while stump-sprout leaves can be exaggerated. Scarlet oak overlaps and hybridizes with black and other red oaks, and northern pin oak can be nearly indistinguishable where geography is ignored. Oak wilt, drought scorch, leaf spots, and defoliation alter color and crown density but not identity.
Evidence limit	Deeply cut foliage should trigger a scarlet-pin-northern-pin comparison, not an immediate species call. Require cup depth and scale posture, nut-tip rings when present, lower-leaf hair, buds, bark, and verified regional site. Outside the documented range or without mature fruit, a red-oak-group or explicit complex answer is often more defensible.

Confuser control

Alternative	Visible separator	Resolving view
<i>Quercus velutina</i>	Black oak tends to have densely hairy terminal buds, more diffuse lower-leaf pubescence, yellow inner bark, and loose or ragged cup-rim scales. Scarlet oak more often has clean deeply cut leaves, less hairy buds, tight cup scales, and concentric rings at the nut tip.	Terminal buds, lower leaf surface, cup rim and nut tip, intact mature bark and any already exposed fresh inner bark.
<i>Quercus rubra</i>	Northern red oak usually has shallower sinuses, a much shallower saucer cup, and long smooth bark tracks; scarlet oak usually has narrow lobes separated by deep rounded sinuses and a deeper bowl cup. Individual shade and sprout leaves overlap.	Upper-crown leaves, complete mature acorns in profile, clustered terminal buds, and bark from both lower and upper bole.
<i>Quercus ellipsoidalis</i> or <i>Quercus palustris</i>	Northern pin oak is centered on western Great Lakes xeric sands and often has an ellipsoid nut in a deep cup; pin oak is a wet-flat species with a shallow cup and strongly tiered crown. Scarlet oak is eastern and dry-ridge centered, but leaves alone may be inseparable.	Mature acorn and cup, georeferenced location, soil drainage and landform, lower-crown architecture, and multiple leaves.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Intermediate to intolerant and strongly light-demanding for successful recruitment; suppressed seedlings persist poorly under a closed canopy.
Moisture / soils	Most characteristic of dry, acidic, well-drained sandy or rocky soils on ridges and upper slopes; growth improves on deeper soils but the species avoids wet lowlands.
Geographic affinity	Appalachian and central-eastern oak centered south of the northern-hardwood core, reaching the course region mainly in PA/NJ, southern NY, southern New England, and the southern Great Lakes fringe.
Associations	Occurs with chestnut, black, northern red, and white oaks, hickories, blackgum, red maple, pitch or white pine, sassafras, and ericaceous shrubs.
Regeneration	Animal-dispersed acorns and abundant advance seedlings or sprouts recruit after canopy opening; low seedling height and competing shrubs often limit regeneration despite adequate acorn crops.
Vegetative reproduction	Seedlings, saplings, and cut stems sprout vigorously from the root collar or stump, allowing persistence after top-kill and supporting coppice-origin stands.
Seed / mast	Acorns mature in the second year; good crops are periodic rather than annual, and weevils, squirrels, deer, birds, and fungi remove many fruits.
Establishment	Thin litter or mineral-soil contact, adequate first-season moisture, and timely high light are needed; prescribed disturbance is effective only where advance reproduction, seed source, and competition are addressed.
Succession	Fire-associated, disturbance-dependent mid-successional oak that can dominate dry ridges but declines in regeneration under prolonged canopy closure and mesophication.
Disturbance response	Mature bark provides some fire resistance and smaller stems resprout after top-kill. Repeated moderate fire can maintain open oak systems, while intense fire, logging injury, or compounded drought and defoliation can cause mortality.
Longevity / growth	Moderately fast when released, commonly maturing within roughly 80-150 years and generally shorter-lived than white or northern red oak.
Browse	Deer consume acorns and browse seedlings and sprouts; repeated browse can remove the height advantage created by cutting or fire.
Insects / disease	Highly susceptible to oak wilt as a red-oak-group species; spongy moth, two-lined chestnut borer, acorn weevils, cankers, and root rots become especially damaging after drought or defoliation.
Management	Maintain adequate light, large advance reproduction, seed trees, and sprout sources on appropriate dry sites; coordinate fire or harvest with browse and competitor control, and diagnose declining crowns rather than treating red foliage as identification.
Silvicultural systems	Shelterwood, group or even-aged regeneration methods, and carefully prescribed fire can favor established reproduction and sprouts; repeated light selection without understory treatment usually shifts composition toward tolerant competitors.

Personal field cue for Scarlet oak

learner-created retrieval hook

Northern pin oak - *Quercus ellipsoidalis*

New this week | Family: Fagaceae | Tier B | Priority 2 - regional

MN/WI/MI and western Great Lakes sand country; rare/local in southern ON

Fire-associated oak of dry nutrient-poor sands; hybridizes with red oaks



Bark

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Dormant

(c) Dominique Menard, some rights reserved...
INAT_PHOTO_458330876 | cc-by



Foliage

(c) Alaina Larkin, some rights reserved...
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Evidence	Diagnostic interpretation
Foliage	Alternate leaves commonly have 5-7 narrow bristle-tipped lobes separated by deep rounded sinuses, with a mostly hairless lower surface except small axillary tufts. Shape overlaps scarlet and pin oaks and varies on sun, shade, and sprout shoots, so a leaf-only identification is rarely defensible.
Bark age series	Young bark is smooth gray-brown; older stems develop shallow fissures and irregular dark gray to blackish ridges, often with persistent dead branch stubs. Bark can become blocky but lacks a unique field character that consistently separates it from black, scarlet, or pin oak.
Dormant	Reddish-brown twigs end in a clustered group of small ovoid to conic reddish-brown buds, hairless to lightly hairy. Buds establish the red-oak group but overlap scarlet and pin oak; densely woolly buds instead favor black oak.
Fruit / cone / seed	Acorns mature in two seasons. The nut is often distinctly ellipsoid, about 12-20 mm long, with a bowl-shaped cup covering roughly one-third to one-half and bearing closely appressed scales. Both nut shape and cup depth vary, especially in introgressed populations, so examine several mature fruits from one crown.
Whole tree / context	A small to medium canopy tree with an irregular crown and retained dead lower branches, most characteristic of dry acidic nutrient-poor sand plains, dunes, outwash, and sandstone hills in MN/WI/MI. Extensive local stands on xeric sands are much stronger context than a planted individual.
Variation / condition	Northern pin oak intergrades and hybridizes with northern red and black oak, and its relationship to scarlet oak has a history of alternative treatment. Leaves and acorns may be intermediate within mixed stands. Fire scars, drought dieback, oak wilt, and retained dead limbs alter condition but are not identity characters.
Evidence limit	A reliable field determination needs a western Great Lakes xeric-sand context plus a series of deep-lobed leaves, mature ellipsoid acorns with deep cups, buds, and bark. Range and soil cannot rescue a contradictory specimen, and morphology cannot always resolve introgression; use a red-oak-group complex or note probable hybrid when characters conflict.

Confuser control

Alternative	Visible separator	Resolving view
<i>Quercus palustris</i>	Pin oak is native chiefly to wet southern lowlands and flats, has a small nearly round nut in a shallow saucer cup covering about one-quarter, and often shows strongly descending lower branches. Northern pin oak occupies dry western Great Lakes sands and usually has an ellipsoid nut in a deeper cup.	Mature acorn in side view, whole crown including lower branches, georeferenced location, and direct soil-drainage or landform evidence.
<i>Quercus coccinea</i>	Scarlet oak is centered farther east and south on dry acidic ridges and often shows a tight cup plus concentric rings around the nut tip; northern pin oak is Great Lakes-centered, commonly has ellipsoid nuts, and is strongly tied to xeric sand. Leaves are often not separable.	Several mature acorns showing the nut tip and cup, verified locality, soil parent material, terminal buds, and population-level variation.
<i>Quercus rubra</i> or <i>Quercus velutina</i>	Northern red oak usually has shallower sinuses and a shallow saucer cup; black oak more often has hairy buds, lower-leaf pubescence, and yellow inner bark. Northern pin oak tends to deep sinuses, a deeper cup, ellipsoid nut, and xeric-sand context, but hybridization is common.	Terminal buds, lower leaf surface, several mature fruits, upper-bole bark, soil context, and neighboring oak morphology.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Very intolerant of shade; it does not regenerate successfully beneath its own closed canopy and requires substantial light.
Moisture / soils	Specialized on dry acidic, nutrient-poor sands and sandy or sandstone uplands with a thin organic layer; highly drought tolerant and absent from truly wet pin-oak sites.
Geographic affinity	A north-central oak with its core-region center in MN, WI, and MI; occurrence far east of that core demands careful documentation of planting, range, and identity.
Associations	Forms pure or mixed stands with northern red, black, white, bur, and sometimes scarlet oak, plus jack or white pine, hophornbeam, hickories, aspen, and barrens shrubs.
Regeneration	Acorns dispersed by gravity, squirrels, and birds establish seedlings after disturbance; open conditions and reduced competition are essential.
Vegetative reproduction	Root collars and stumps sprout vigorously following cutting, fire top-kill, or crown injury, often sustaining stands where seedling recruitment is episodic.
Seed / mast	Second-season acorns begin on relatively young trees; good crops are intermittent, with substantial losses to weevils and wildlife.
Establishment	Exposed or lightly covered sandy seedbeds, adequate early moisture, and full or nearly full light favor recruitment; litter, shade, browse, and pine or shrub competition can prevent establishment.
Succession	Facultative-seral and fire-associated; on extremely poor sands it may act as an edaphic climax, while on better sites it is succeeded by less light-demanding oaks, white pine, or mesic hardwoods.
Disturbance response	Mature bark can withstand low-intensity fire and top-killed small stems sprout strongly. Fire often maintains open habitat, although frequent severe burning can reduce small-stem populations.
Longevity / growth	Slow to moderate growth on infertile sands, usually a small to medium tree with drought-adapted roots rather than a high-yield timber form.
Browse	Acorns and young shoots are eaten by deer, squirrels, jays, and other wildlife; repeated deer browse can delay sprout and seedling recruitment.
Insects / disease	Susceptible to oak wilt and common oak defoliators, borers, galls, and acorn insects; condition symptoms overlap other red oaks and must not be used as an identity shortcut.
Management	Conserve xeric oak barrens and verified local seed sources, retain sprout capacity and adequate seed trees, use range-aware identifications, and label morphologically intermediate stems explicitly rather than assuming pure northern pin oak.
Silvicultural systems	Clearcut or shelterwood systems can regenerate stands when advance reproduction is adequate or deliberately established; prescribed fire and competitor control can maintain open conditions, while partial shade without follow-up release favors succession away from the species.

Personal field cue for Northern pin oak

learner-created retrieval hook

Pin oak - *Quercus palustris*

New this week | Family: Fagaceae | Tier B | Priority 2 - regional

Wet bottomlands in the southern Great Lakes and southern ON east through NY, PA/NJ, CT, and MA; widely planted

Shade-intolerant wet-site red-oak-group tree and common planted confuser



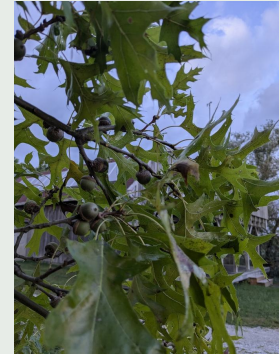
Bark

(c) Ben Redding, some rights reserved...
INAT_PHOTO_372982628 | cc-by



Dormant

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INAT_PHOTO_553932678 | cc0



Mixed Evidence

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Evidence	Diagnostic interpretation
Foliage	Alternate leaves usually have 5-7 narrow bristle-tipped lobes divided by deep broad U-shaped sinuses. The lower surface is pale with small rusty tufts in vein axils. Foliage closely overlaps scarlet and northern pin oak and must be joined to acorn, crown, and site evidence.
Bark age series	Young and pole-size trees retain thin, smooth gray-brown bark for many years. Older boles become shallowly ridged and narrowly fissured but are usually less thick and blocky than mature black or northern pin oak. Trunks often retain many small dead branch stubs that produce the characteristic pin knots.
Dormant	Slender reddish-brown twigs end in clusters of small pointed chestnut-brown buds that are usually hairless to only finely hairy. Numerous persistent short branchlets and dead lower branches support the whole-tree pattern but do not distinguish a detached twig.
Fruit / cone / seed	Acorns mature in two seasons and are small, about 10-15 mm, nearly hemispherical to rounded, often faintly striped, and seated in a thin shallow saucer cup covering about one-quarter or less of the nut. This shallow cup is a critical contrast with northern pin and scarlet oak.
Whole tree / context	Open-grown trees have a strongly excurrent, pyramidal crown: upper branches ascend, middle branches spread horizontally, and lower branches droop. Native trees occupy seasonally wet poorly drained flats and bottomlands; planted street and lawn trees occur widely outside that context.
Variation / condition	Forest crowding can obscure the tiered crown, pruning removes lower branches, and planted trees may be far outside the native range. Alkaline soils often cause interveinal chlorosis, while oak wilt, bacterial leaf scorch, drought, and defoliation cause additional discoloration. None of these conditions proves species identity.
Evidence limit	Deep-lobed leaves plus a pyramidal crown are suggestive, not sufficient. A species-level call should include the small round acorn with shallow cup and either authentic wet-flat ecology or documented planting. Without fruit and context, answer the deep-lobed red-oak complex; diagnose chlorosis or wilt separately.

Confuser control

Alternative	Visible separator	Resolving view
<i>Quercus ellipsoidalis</i>	Northern pin oak is a xeric western Great Lakes sand species and usually has an ellipsoid nut in a cup enclosing one-third to one-half. Pin oak is naturally a wet-flat tree with a small round nut in a shallow cup and often a strongly tiered crown.	Mature fruit in side view, whole crown including lower limbs, georeferenced location, and evidence of soil drainage or planting.
<i>Quercus coccinea</i>	Scarlet oak is associated with dry acidic ridges and has a deeper bowl cup, often with concentric rings at the nut tip. Pin oak occurs naturally on poorly drained flats, has a shallow cup, and commonly retains drooping lower branches; leaves alone overlap strongly.	Complete mature acorn, slope and drainage context, lower-crown architecture, terminal buds, and multiple upper-crown leaves.
<i>Quercus velutina</i>	Black oak more often has densely hairy terminal buds, lower-leaf pubescence, yellow inner bark, blocky mature bark, and a deeper loose-scaled cup. Pin oak has small smoother buds, a small shallow-cupped fruit, smoother bark, and wet-flat context.	Terminal buds, lower leaf surface, mature cup, mature bole bark, and site drainage.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Intolerant of shade; overtopped seedlings and intermediates decline quickly, and natural stands are mainly even-aged dominants or codominants.
Moisture / soils	Characteristic of poorly drained, seasonally flooded heavy clays, glacial till flats, and bottomlands; it tolerates dormant-season inundation but not permanent deep flooding and often chloroses on alkaline uplands.
Geographic affinity	Central lowland species reaching the course region in southern Great Lakes and northeastern bottomlands; widely planted north and east of native populations.
Associations	Associates with swamp white oak, silver and red maples, American elm, green ash, cottonwood, sycamore, blackgum, and other wet-bottomland hardwoods.
Regeneration	Abundant animal- and gravity-dispersed acorns can create dense reproduction on suitable wet flats; rapid release is necessary because seedlings do not tolerate overtopping.
Vegetative reproduction	Young stumps and injured seedlings sprout vigorously from dormant stem or root-collar buds; sprouting declines with age and large size.
Seed / mast	Second-season acorns are produced frequently after maturity, with good crops in many years; wildlife and acorn insects remove large quantities.
Establishment	Moist exposed soil or thin litter on seasonally wet but aerated microsites and high light after germination; prolonged seedling submergence or dense shade causes failure.
Succession	Subclimax and disturbance-dependent on heavy wet soils, where abundant regeneration can persist if released; on better-drained sites it is displaced by more tolerant hardwoods.
Disturbance response	Young trees sprout after top injury and established trees tolerate periodic flooding, but thin bark is fire-sensitive. Drainage change, permanent flooding, severe soil disturbance, and alkalinity can cause decline.
Longevity / growth	Rapid juvenile growth on suitable bottomlands but comparatively short-lived for an oak, often reaching physiological maturity near 80-100 years and rarely forming ancient trees.
Browse	Deer and other wildlife consume acorns and browse young stems; heavy browse can suppress otherwise abundant regeneration.
Insects / disease	Highly susceptible to oak wilt; common oak defoliators, borers, galls, acorn weevils, bacterial leaf scorch, and nutrient-related chlorosis also occur. Condition patterns are diagnostically separate from identity.
Management	Reserve it for appropriate wet mineral-soil sites, protect hydrology, release adequate reproduction promptly, avoid moving planted-tree assumptions into native inventories, and distinguish nutrient chlorosis from lethal vascular disease.
Silvicultural systems	Even-aged regeneration or clearcutting can work where abundant advance reproduction is present on suitable flats; small light openings and selection are poor matches for this shade-intolerant species, and hydrologic damage can negate any silvicultural treatment.

Personal field cue for Pin oak

learner-created retrieval hook

White oak - *Quercus alba*

New this week | Family: Fagaceae | Tier A | Priority 1 - core

Broad in southern and western portions of the region, north locally to southern Canada and New England

Long-lived mast and timber tree; central white-oak-group reference



Bark

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Foliage

(c) Rose Zappa, some rights reserved...
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Reproductive

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Evidence	Diagnostic interpretation
Foliage	Alternate leaves have 7-9 rounded, bristleless lobes, with sinuses ranging from shallow to deep and an often wedge-shaped base. The upper surface is blue-green to dark green and the lower surface paler, usually becoming nearly hairless. Confirm multiple mature leaves because shade, sun, epicormic, and juvenile foliage differ markedly.
Bark age series	Young stems are smooth light gray to brownish gray. Pole and mature bark becomes pale ashy gray and breaks into irregular overlapping flaky plates or narrow loose scales; very old lower boles can become deeply furrowed and darker while upper bark retains paler scaling.
Dormant	Gray to reddish twigs terminate in a tight cluster of short, blunt to ovoid reddish-brown, mostly hairless buds. Alternate leaf scars and clustered buds identify oak; short rounded buds and pale twigs support the white-oak group but overlap swamp white and bur oak.
Fruit / cone / seed	White-oak-group acorns mature in one growing season. The oblong to ovoid nut, commonly 15-25 mm long, is enclosed about one-quarter by a shallow bowl cup with warty, closely appressed scales and is sessile or on a short stalk. Sound acorns commonly begin root germination soon after autumn fall.
Whole tree / context	A large, long-lived tree with a broad rounded crown and massive limbs when open-grown, occurring on well-drained uplands from dry ridges to productive loams. Within the course region it is strongest south and west of the northern-hardwood center and may be local near its northern limits.
Variation / condition	Leaf depth, bark plating, acorn size, and form vary among individuals and sites, and white oak hybridizes with several members of the white-oak group. Wet or lichen-covered bark can appear darker; fire scars and old wounds alter the lower bole. Oak wilt can occur but typically progresses differently than in red oaks and is condition evidence only.
Evidence limit	Rounded bristleless lobes place a specimen in the white-oak group but do not alone prove <i>Q. alba</i> . A strong field call combines several leaves, pale flaky mature bark, clustered short buds, shallow short-stalked cup, and upland range/site. Young bark, winter twigs, or an isolated fallen acorn without source-tree association may require a white-oak-group answer.

Confuser control

Alternative	Visible separator	Resolving view
<i>Quercus bicolor</i>	Swamp white oak usually has shallow irregular lobes or coarse rounded teeth, a densely white felted lower surface, and acorns on a conspicuously long common stalk. White oak usually has more definite rounded lobes, a less densely hairy underside, and sessile or short-stalked acorns.	Lower leaf surface, complete fruit stalk with attachment, several mature leaves, and direct evidence of upland versus seasonally flooded habitat.
<i>Quercus macrocarpa</i>	Bur oak often has a very large leaf with a deep middle waist, corky twig ridges, thick dark furrowed bark, and a large acorn deeply enclosed by a mossy-fringed cup. White oak lacks the massive fringe and usually has paler flaky bark and a shallow cup.	Complete leaf with scale, twig internodes, mature cup in side view, and mature bark above any scarred base.
<i>Quercus montana</i>	Chestnut oak leaves are unlobed but have coarse rounded chestnut-like teeth, mature bark is dark and deeply blocky, and the acorn cup is deeper. White oak has true rounded lobes and usually pale flaky bark.	Several complete leaves including their bases, mature bark, clustered buds, and mature acorn and cup.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Intermediate and generally more shade tolerant as a seedling than the most light-demanding red oaks, but sustained canopy recruitment requires substantial release.
Moisture / soils	Occupies a broad range from dry ridges to moist well-drained coves, growing best on deep fertile loams and clay loams while avoiding permanent saturation.
Geographic affinity	Broad eastern and central species, common in the southern and western course region and increasingly local toward northern New England, New Brunswick, Quebec, and northern Ontario.
Associations	Occurs with northern red, black, scarlet, chestnut, bur, and swamp white oaks, hickories, red maple, blackgum, white pine, tulip tree, and northern hardwoods on favorable sites.
Regeneration	Animal-cached annual acorns produce seedlings and advance reproduction; successful replacement depends on sufficient seedling size, timely light, browse control, and competition reduction.
Vegetative reproduction	Seedlings and young stems sprout from root collars or stumps after fire, cutting, or top-kill; large old trees have reduced sprouting capacity.
Seed / mast	One-season acorns are produced in variable crops, with heavy mast separated by poor years; they germinate quickly in autumn and are highly preferred by wildlife and acorn insects.
Establishment	Fresh sound acorns, moist seedbeds with limited litter barriers, and protection from predators support establishment; seedlings can begin under partial shade but need release before tolerant competitors overtop them.
Succession	Long-lived mid- to late-successional canopy species that can persist through repeated disturbance, yet often fails to replace itself under dense shade and fire exclusion.
Disturbance response	Thickening bark and strong sprouting confer fire resilience with size; moderate fire or harvest can release regeneration, while severe fire, root injury, compaction, and compounded drought and defoliation cause decline.
Longevity / growth	Slow to moderate juvenile growth, strong long-term persistence, and potential life spans of several centuries; invests in durable wood and crown longevity rather than rapid pioneer growth.
Browse	Acorns are highly preferred and seedlings and sprouts are browsed by deer and other mammals; repeated browse can eliminate the larger advance reproduction needed for regeneration.
Insects / disease	Oak wilt can affect white oak but is often slower than in red-oak-group trees; spongy moth, two-lined chestnut borer, acorn weevils, anthracnose, root rots, and decline complexes are important under stress.
Management	Protect long-lived seed and crop trees, maintain large advance reproduction and sprout sources, coordinate harvest or fire with competitor and browse control, and preserve genetically and structurally diverse white-oak-group components.
Silvicultural systems	Shelterwood and group or even-aged systems that first develop advance reproduction and then release it are reliable; prescribed fire can improve competitive position on suitable sites, whereas repeated light selection generally favors shade-tolerant mesophytes.

Personal field cue for White oak

learner-created retrieval hook

Swamp white oak - *Quercus bicolor*

New this week | Family: Fagaceae | Tier B | Priority 2 - regional

Southern Great Lakes and lower Great Lakes through NY, PA/NJ, and southern New England

Flood-tolerant white-oak-group tree of swamps, terraces, and floodplains



Bark

(c) Peter Quakenbush, some rights reserved...
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Foliage

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Mixed Evidence

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Evidence	Diagnostic interpretation
Foliage	Alternate obovate leaves have shallow irregular rounded lobes or coarse rounded teeth and a wedge-shaped base. The dark green upper surface contrasts strongly with the densely white to gray felted lower surface, producing the 'bicolor' effect; the petiole is relatively short. Confirm hair rather than relying on reflected light.
Bark age series	Young bark is smooth gray-brown. Older bark becomes gray, scaly, and irregularly flaky, often peeling in strips or plates and resembling white oak; branch bark may be especially shaggy. Mature bark is supportive only when paired with foliage and fruit.
Dormant	Yellow-brown to gray-brown twigs carry clustered short blunt buds; young twigs may be somewhat hairy. Dormant material overlaps white and bur oak, so seek persistent long fruit stalks, shaggy branch bark, and lowland context rather than forcing a twig-only species call.
Fruit / cone / seed	Acorns mature in one season, commonly one or two together on a conspicuously long peduncle 25-75 mm or more. The ovoid nut is enclosed about one-third by a deep bowl cup whose upper scales may be thickened, pointed, or weakly fringed. Retain the full stalk in specimens because detached cups lose the strongest character.
Whole tree / context	A medium to large lowland oak with a broad irregular crown, occurring on stream terraces, swamp margins, depressions, and bottomlands that flood periodically but are not permanently inundated. It can stand on slight rises within wet complexes and therefore need not be rooted in visible water.
Variation / condition	Leaf lobing and lower-surface hair density vary, and fruit stalk length is easier to assess on intact clusters than fallen acorns. Swamp white oak hybridizes with bur and white oak. Flood scars, epicormic shoots, wet bark, and oak wilt or scorch affect condition but are not species characters.
Evidence limit	The strongest combination is a white felted lower leaf surface, long-stalked one-season acorns, flaky gray bark, and periodically flooded habitat. A pale underside alone can be confused with bur oak, and bark or buds alone rarely resolve the species. If fruit attachment and site are unavailable, retain a white-oak-group identification.

Confuser control

Alternative	Visible separator	Resolving view
<i>Quercus alba</i>	White oak usually has more definite rounded lobes, a lower surface that becomes less densely felted, and sessile or short-stalked shallow-cupped acorns. Swamp white oak has a striking white underside and one or two deeper-cupped acorns on a long common stalk.	Lower leaf surface at high detail, intact fruiting peduncle, several leaves, bark above breast height, and site drainage.
<i>Quercus macrocarpa</i>	Bur oak can also have pale leaf undersides but often shows a deep central leaf waist, corky twig wings, thick deeply furrowed bark, and a very large cup with a conspicuous mossy fringe. Swamp white oak usually has a long slender fruit stalk and a less massive cup.	Complete leaves, twig internodes, fruit stalk and cup profile, and mature trunk bark.
<i>Quercus montana</i>	Chestnut oak has coarsely rounded teeth but its lower leaf surface is not densely white felted, its fruit is short-stalked, and its habitat and mature bark are dry-ridge and deeply blocky rather than wet-lowland and flaky.	Lower leaf surface, complete acorn stalk, mature bole bark, and landform or hydrology view.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Intermediate; seedlings can establish under moderate shade but require release for sustained growth and canopy recruitment.
Moisture / soils	Most common on hydromorphic mineral, alluvial, or organic soils with imperfect to poor drainage and periodic flooding, but absent from permanently inundated sites.
Geographic affinity	A north-central and northeastern lowland oak strongest from the southern Great Lakes through lower Great Lakes and northeastern floodplains.
Associations	Associates with pin and bur oak, red and silver maple, American elm, green ash, cottonwood, sycamore, blackgum, black willow, and basswood.
Regeneration	One-season acorns dispersed by gravity, water, birds, and mammals establish on moist aerated microsites; seedlings need release from shade and flood-tolerant competition.
Vegetative reproduction	Young stems and seedlings can sprout after cutting or injury, but natural regeneration planning should not assume the vigorous coppice response of every upland oak.
Seed / mast	Acorns ripen in autumn on long stalks, with crop size varying among years; ducks, deer, squirrels, and other wildlife consume and disperse the mast.
Establishment	Moist exposed soil on hummocks, terraces, or better-aerated lowland microsites, protection from prolonged submergence, and adequate light following establishment.
Succession	A persistent lowland associate that may become dominant on better-drained wetland soils; flood variability makes succession less predictable than on uplands.
Disturbance response	Tolerates periodic flooding after establishment but not permanent saturation, severe root disturbance, or repeated fire. Canopy opening releases seedlings, while hydrologic alteration can rapidly change suitability.
Longevity / growth	Moderately fast on good lowlands and long-lived, often reaching 300 years or more, with shallow roots adapted to high water tables.
Browse	Acorns and young shoots are eaten by deer and other wildlife; lowland browse concentration can suppress regeneration.
Insects / disease	Subject to oak wilt, anthracnose, leaf spots, borers, galls, root rots, and acorn insects, but no condition symptom is diagnostic for the species.
Management	Protect natural flood regimes and soil structure, retain seed trees and aerated microsites, account for browse, and avoid substituting the species indiscriminately on permanently wet or alkaline disturbed ground.
Silvicultural systems	Even-aged openings, including clearcutting on suitable better-drained lowland sites, can release abundant reproduction; any harvest system must minimize rutting and preserve hydrology, and partial cuts succeed only with adequate light and established seedlings.

Personal field cue for Swamp white oak

learner-created retrieval hook

Bur oak - *Quercus macrocarpa*

New this week | Family: Fagaceae | Tier B | Priority 2 - regional

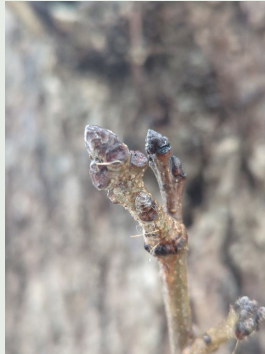
Strongest in MN/WI/MI and southern ON; local eastward on calcareous or floodplain sites

Fire- and drought-tolerant savanna/woodland oak with thick bark and large mast



Bark

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Dormant

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Foliage

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Evidence	Diagnostic interpretation
Foliage	Alternate leaves are large and obovate with rounded bristleless lobes, commonly narrowed by a pair of very deep middle sinuses into a distinctive waist and expanded into a broad terminal lobe. The lower surface is pale and variably hairy. Examine multiple leaves because some lack the textbook waist.
Bark age series	Young bark is gray and becomes corky at an early age. Mature boles have very thick dark gray-brown bark divided into deep furrows and heavy rugged ridges, an adaptation associated with fire-prone savannas. Old open-grown trees often carry massive rough limbs.
Dormant	Stout gray-brown twigs end in clustered short rounded buds and commonly develop conspicuous corky ridges or wings along internodes. Corkiness varies and may be absent on a sampled twig, but when combined with stoutness and the bud cluster it strongly supports bur oak.
Fruit / cone / seed	One-season acorns are among the largest regional oak fruits, commonly 25-50 mm. A deep thick cup encloses one-half or more of the nut and usually bears a conspicuous mossy fringe of elongated scales around the rim. Northern populations can have smaller, thinner, less strongly fringed cups, so assess a series.
Whole tree / context	A rugged, broad-crowned tree of prairie-forest margins, oak openings, dry sandy or limestone uplands, savannas, and also elevated terraces or ridges in floodplains. It is strongest in the western Great Lakes and southern Ontario and becomes local eastward.
Variation / condition	Leaf waists, twig cork, acorn size, and cup fringe vary strongly, including a recognized northern fruit form, and bur oak hybridizes with swamp white and white oak. Browse, repeated fire, open-grown architecture, and flood or mechanical scars change form without changing identity.
Evidence limit	The deep fringed cup is the best single structure, but northern or hybrid fruits may be less extreme. Combine fruit, large variably waisted leaves, corky twigs, thick bark, and regional site. A white-oak-group leaf or deeply furrowed old bole alone is insufficient, and apparent hybrids should be recorded as such or left in a species complex.

Confuser control

Alternative	Visible separator	Resolving view
<i>Quercus bicolor</i>	Swamp white oak has a densely white felted leaf underside and acorns on a conspicuously long slender stalk, usually with a less massive cup. Bur oak more often has a deeply waisted leaf, corky twigs, thick bark, and a large deeply enclosing mossy-fringed cup.	Full fruit stalk and cup, lower leaf surface, twig internodes, several leaves, and mature trunk bark.
<i>Quercus alba</i>	White oak usually has several more evenly spaced rounded lobes, pale flaky bark, non-corky twigs, and a smaller shallow cup without a mossy fringe. Bur oak is coarser in leaf, twig, bark, and fruit, but hybrids can be intermediate.	Several mature leaves, twigs, cup in profile, upper-bole bark, and population context.
<i>Quercus montana</i>	Chestnut oak leaves are unlobed with coarse rounded teeth, its cup lacks a mossy fringe, and it is centered on dry Appalachian ridges. Bur oak has true rounded lobes, often a deep waist, a fringed cup, corky twigs, and a western Great Lakes-prairie affinity.	Complete leaf margins, cup rim, twig corkiness, verified location, and landscape context.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Intermediate but unable to recruit indefinitely under dense shade; seedlings and saplings need release, especially where maple and basswood are advancing.
Moisture / soils	Exceptionally broad, from droughty sands, thin limestone ridges, claypans, and prairie loams to moist elevated bottomland terraces; avoids long permanent inundation.
Geographic affinity	Continental and cold- and drought-tolerant, with greatest course-region abundance in MN/WI/MI and southern Ontario and scattered calcareous or floodplain occurrences eastward.
Associations	Dominates oak savannas and openings and associates with northern pin, white, black, and swamp white oaks, basswood, sugar maple, green ash, elm, bitternut hickory, and prairie-edge shrubs.
Regeneration	Large acorns dispersed by gravity and animals establish seedlings on open mineral or lightly littered soil; advance reproduction and sprouts need adequate light to remain competitive.
Vegetative reproduction	Young trees and seedlings sprout after cutting or fire top-kill, while sprouting capacity declines with age and diameter.
Seed / mast	One-season acorns vary strongly in size and crop among years and regions; northern forms may delay germination until spring, while typical fruits may germinate in autumn. Wildlife consumes much of the mast.
Establishment	Open or partially open sites with adequate first-year moisture, exposed soil, and reduced grass, shrub, or tolerant-tree competition; cached acorns aid establishment beyond crown drip lines.
Succession	Pioneer and persistent canopy species on prairie margins and severe dry sites; on mesic fire-protected sites it is commonly succeeded by shade-tolerant maple-basswood or beech-maple vegetation.
Disturbance response	Thick bark makes large trees highly resistant to low-intensity fire and young top-killed stems can sprout. Fire exclusion promotes mesophication, while severe repeated fire can still damage regeneration and roots.
Longevity / growth	Slow on harsh uplands and moderate on fertile bottomlands, often living 200-300 years or more and developing massive crowns and boles.
Browse	Large acorns are valuable mast and young shoots are browsed by deer and livestock; savanna regeneration can be lost to combined browse and grass competition.
Insects / disease	Oak wilt can affect bur oak, as can bur oak blight in parts of the upper Midwest, oak decline, root rots, borers, defoliators, cankers, and acorn insects. Symptoms require independent diagnosis.
Management	Retain old fire-resistant seed trees and locally adapted seed sources, restore adequate light and disturbance where savanna is the objective, protect regeneration from browse and competing sod, and preserve hydrology on bottomland sites.
Silvicultural systems	Shelterwood, group openings, coppice contributions, and prescribed fire can regenerate bur oak when timed to advance reproduction and competitor control; repeated light selection and fire exclusion tend to favor shade-tolerant associates.

Personal field cue for Bur oak

learner-created retrieval hook

Chestnut oak - *Quercus montana*

New this week | Family: Fagaceae | Tier B | Priority 2 - regional

Allegheny Plateau and ridges of PA/NJ/NY and southern New England; isolated westward

Dry-ridge white-oak-group dominant with strong sprouting



Bark

(c) Bonnie Semmling, some rights...
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Dormant

(c) Alex Zorach, some rights reserved...
INAT_PHOTO_342763967 | cc-by



Foliage

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Evidence	Diagnostic interpretation
Foliage	Alternate obovate to elliptic leaves are unlobed but have 10-16 pairs of large rounded crenate teeth, resembling a chestnut leaf with blunt rather than sharp teeth. The upper surface is yellow-green and the lower surface paler and variably hairy, especially when young.
Bark age series	Young bark is smooth gray-brown. Mature bark becomes exceptionally thick, dark gray-brown, and deeply divided by V-shaped furrows into long broad blocky ridges; upper bark can be less extreme. Bark is strong contextual evidence on old ridge trees but overlaps other rough-barked oaks.
Dormant	Stout reddish- to gray-brown twigs end in clustered pointed reddish-brown buds, sometimes finely hairy. Buds and leaf scars establish white-oak-group affinity poorly, so winter identification should add massive ridge bark, retained acorn cups, and dry Appalachian landform.
Fruit / cone / seed	Acorns mature in one season. The shiny chestnut-brown ovoid nut is commonly 20-35 mm long and held about one-third to nearly one-half in a deep hemispheric cup with thick, tightly appressed warty scales; the fruit is sessile or short-stalked and lacks bur oak's fringe.
Whole tree / context	A medium to large tree, often with a crooked bole and irregular crown, dominant on dry rocky acidic ridgetops, upper slopes, and sandstone outcrops of the Allegheny Plateau and Appalachians. It can occupy better sites, but ridge position and thick bark are a powerful regional combination.
Variation / condition	Shade leaves can have shallower teeth, young bark lacks massive blocks, and sprouts carry oversized leaves. The name <i>Q. prinus</i> has been applied inconsistently in older literature, sometimes entangling chestnut and swamp chestnut oaks; the course accepts it for <i>Q. montana</i> only when clearly intended. Fire scars, lichen, and oak wilt are conditions, not identity.
Evidence limit	Rounded chestnut-like teeth plus deeply blocky bark and dry Allegheny/Appalachian ridge context support species level; add the one-season deep unfringed cup whenever possible. A detached unlobed leaf may be American chestnut or another chestnut-leaved oak, and a young twig alone should remain white-oak group.

Confuser control

Alternative	Visible separator	Resolving view
<i>Castanea dentata</i>	American chestnut leaves are longer and more lanceolate with many sharp, forward-pointing hooked teeth, while chestnut oak has fewer large rounded teeth. Chestnut bears spiny burs rather than acorns and lacks clustered oak terminal buds.	Leaf margin and base in scale, terminal buds, any fruit or cup, and mature bark.
<i>Quercus bicolor</i>	Swamp white oak has a densely white felted lower leaf surface, long-stalked acorns, flaky gray bark, and lowland flooding context. Chestnut oak has a less white underside, short-stalked deeper cups, massive dark blocky bark, and dry ridge context.	Lower leaf surface, complete fruit stalk, mature bark, and a landscape view establishing ridge versus lowland position.
<i>Quercus alba</i> or <i>Quercus macrocarpa</i>	White and bur oak leaves have true rounded lobes; bur oak also has a mossy-fringed cup and often corky twigs, while white oak has pale flaky bark and a shallow cup. Chestnut oak is unlobed with rounded teeth and has a deep unfringed cup.	Several complete leaves, cup rim and profile, twig internodes, and mature upper-bole bark.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Intermediate; seedlings may persist beneath an open canopy but successful recruitment depends on substantial light and release.
Moisture / soils	Most characteristic of dry, rocky, well-drained acidic sandstone and shale ridges with low fertility, though best growth occurs on deeper protected slopes.
Geographic affinity	Appalachian and Allegheny species reaching the course chiefly in PA/NJ, southern NY and New England, with scattered western outliers.
Associations	Associates with scarlet, black, white, and northern red oak, hickories, blackgum, red maple, sweet birch, white or pitch pine, mountain-laurel, and blueberries.
Regeneration	Animal-dispersed acorns establish seedlings and advance reproduction; vigorous sprouts following harvest or top-kill are a major stand-replacement pathway.
Vegetative reproduction	One of the strongest eastern oak sprouters, producing vigorous root-collar and stump sprouts after cutting or fire, especially from young and moderate-size stems.
Seed / mast	One-season acorns are produced in variable crops and dispersed rapidly by gravity and wildlife; predation and acorn weevils create large losses.
Establishment	Thin litter or exposed mineral soil, adequate moisture during germination, and high light after establishment; ericaceous shrubs, tolerant hardwoods, browse, and drought can suppress seedlings.
Succession	Persistent edaphic dominant on dry ridges and a disturbance-dependent mid-successional oak on better sites; fire exclusion can shift mixed stands toward red maple and other mesophytes.
Disturbance response	Thick bark and vigorous sprouting confer resilience to surface fire and cutting, but small stems may be top-killed and repeated severe burning can exhaust sprouts. Drought tolerance is high relative to mesic associates.
Longevity / growth	Slow on harsh ridges, moderate on better slopes, and capable of living several centuries; prioritizes persistence and sprouting over rapid early height growth.
Browse	Acorns and sprouts are heavily used by deer and other wildlife; repeated browsing can hold advance reproduction below competitive height.
Insects / disease	Oak wilt is possible but white-oak-group trees often decline more slowly than red oaks; spongy moth, borers, chestnut oak decline, root rots, and acorn insects are important under stress.
Management	Match the species to dry-ridge objectives, retain vigorous seed trees and sprout sources, control browse and competing ericaceous or mesophytic layers, and treat historical <i>Q. prinus</i> records cautiously where taxonomy is ambiguous.
Silvicultural systems	Shelterwood, group openings, clearcutting with reliable sprout potential, and prescribed fire can sustain chestnut oak; single-tree selection without adequate understory light generally favors more tolerant species.

Personal field cue for Chestnut oak

learner-created retrieval hook

Common hackberry - *Celtis occidentalis*

New this week | Family: Cannabaceae | Tier B | Priority 2 - regional

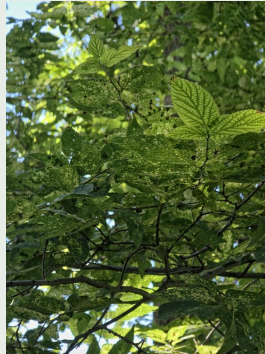
MN/WI/MI and southern ON through NY, PA/NJ, and southern New England

Tough tree of rich bottomlands, limestone sites, and urban settings



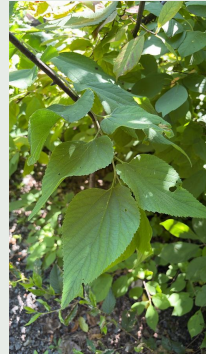
Bark

(c) Sandy Wolkenberg, some rights...
INAT_PHOTO_592707972 | cc-by



Condition

(c) Steven J. Røste, some rights...
INAT_PHOTO_718580407 | cc-by



Foliage

(c) Marc-Olivier Beausoleil, some...
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Evidence	Diagnostic interpretation
Foliage	Alternate simple ovate leaves have a strongly unequal, oblique base, three prominent veins arising near the base, and a serrate margin mainly beyond the lower third. The upper surface is distinctly rough like sandpaper while the lower surface is smoother to softly hairy. Twig attachment must be visible because detached leaves can mimic elm.
Bark age series	Young bark is smooth gray-brown with lenticels. Intermediate stems develop isolated corky warts, and mature gray bark forms distinctive irregular corky knobs, ridges, and projecting plates. Warts may be sparse on vigorous young trees and are not insect galls.
Dormant	Slender alternate twigs are often zigzag, reddish-brown to gray, with small appressed ovoid pointed buds and no true terminal bud; the last lateral bud appears terminal. Leaf scars are small and alternate. Corky bark and persistent fruits greatly improve winter confidence because the twig alone overlaps elm.
Fruit / cone / seed	Small greenish flowers appear with the leaves. Each fruit is a single round drupe about 6-10 mm wide on a long slender drooping pedicel, ripening orange-red to dark purple or nearly black around one hard pitted stone and often persisting into winter. It is not an elm samara or basswood nutlet cluster.
Whole tree / context	A small to medium, occasionally large tree with an irregular rounded crown, common on rich bottomlands, riparian terraces, limestone bluffs, high-pH soils, disturbed edges, and urban sites. It tolerates drought and periodic flooding but not permanent saturation.
Variation / condition	Leaf size, roughness, and bark corkiness vary with age and vigor. Hackberry nipple galls distort leaves, and witches'-broom produces dense twig clusters; both are common conditions but neither is required for identity. Epicormic growth, road salt, drought, and pruning can further alter form.
Evidence limit	Warty corky mature bark can permit rapid field recognition, but young smooth stems require the combined unequal three-veined rough leaf, alternate twig, and long-stalked drupe. A gall-covered detached leaf or zigzag twig alone is insufficient. Sugarberry is mostly outside the course region, but planted or southern material may require a <i>Celtis</i> -level answer.

Confuser control

Alternative	Visible separator	Resolving view
<i>Ulmus</i> species	Elms also have unequal alternate leaves but usually show strongly double-serrate margins, lateral veins running toward each tooth, different buds, and flat samaras. Hackberry has three strong basal veins, a rough upper surface, corky warty bark, and long-stalked round drupes.	Leaf base and venation, complete margin, buds, mature bark, and any fruit or persistent pedicel.
<i>Tilia americana</i>	Basswood leaves are broadly heart-shaped and more nearly symmetrical with palmate veins, and fruits hang as nutlet clusters from a strap-like bract. Hackberry leaves are obliquely based and fruits are solitary long-stalked drupes; mature bark is corky rather than long-ridged.	Complete leaf base, fruit attachment, several twig nodes, and mature bark.
<i>Celtis laevigata</i>	Sugarberry generally has narrower, less rough leaves and smoother bark with fewer prominent corky warts and is principally southern; common hackberry has coarser rough leaves and strongly corky bark. Characters and ranges overlap south of the course envelope.	Multiple mature leaves with texture and measurements, bark from the trunk, fruit, and verified geographic provenance.

Silvics capsule

Dimension	Field-use summary
Shade tolerance	Intermediate to tolerant; seedlings may persist in dense shade, although prolonged suppression produces poor form and release improves growth.
Moisture / soils	Grows on many soils but is strongest on fertile valley alluvium, riparian terraces, and calcareous or high-pH uplands; tolerates drought and periodic flooding but not permanent high water.
Geographic affinity	Broad continental eastern species tolerant of heat, cold, and climatic extremes, especially common in the Great Lakes and central portion of the course region.
Associations	Occurs with elm, green ash, boxelder, silver maple, cottonwood, basswood, sugar maple, bur and swamp white oak, walnut, and other rich-site hardwoods.
Regeneration	Birds and mammals disperse drupes widely; seeds can remain viable in litter through more than one winter, and seedlings establish under existing hardwood stands as well as some disturbed sites.
Vegetative reproduction	Small stumps may sprout and the species can be propagated by cuttings or layering, but large-tree stump sprouting is uncommon and clonal spread is not its main forest strategy.
Seed / mast	Produces good fruit crops in most years with lighter intervening crops; persistent winter drupes are consumed and dispersed by many birds and mammals.
Establishment	Moist but aerated soil, a seed source or animal dispersal, and eventual crown space; young seedlings are more sensitive to saturated soils than mature trees.
Succession	Occurs from temporary to subclimax communities and can enter understories beneath oak or mixed hardwoods; it seldom forms extensive pure forest stands.
Disturbance response	Tolerates urban disturbance, wind, drought, and periodic flooding better than many associates, but thin bark is fire-sensitive and flooding of long duration injures roots and seedlings.
Longevity / growth	Moderate to fast on good alluvium, slow when suppressed or on poor sites, and commonly reaches about 150-200 years as a small to medium canopy tree.
Browse	Deer browse seedlings and twigs, while birds and mammals heavily use the fruit; browsing may limit local recruitment but is not usually the species' sole constraint.
Insects / disease	Nipple galls are conspicuous but usually minor; witches'-broom results from interacting gall mites and powdery mildew. Leaf spots, hackberry engraver, root rot, and decay after fire or injury also occur.
Management	Retain for wildlife fruit, bottomland diversity, and resilience on calcareous sites, protect roots and soil aeration, and do not mistake cosmetic galls or witches'-broom for evidence that a different species is present.
Silvicultural systems	Can persist under selection and respond to moderate release; group openings improve crown form and fruiting, while lowland harvesting must avoid rutting, compaction, and hydrologic change.

Personal field cue for Common hackberry

learner-created retrieval hook

Part II - Cumulative rapid-reference atlas

Use this as a field briefing, not as a substitute for retrieval. Earlier taxa are compressed here so the packet remains self-contained while preserving space for unfamiliar assessment images.

Taxon	High-value visible evidence	Silvics anchor
Sugar maple <i>Acer saccharum</i>	Foliage: Opposite, simple blades are palmately veined and usually five-lobed, with pointed lobes, broad rounded U-shaped... Bark: Seedlings and saplings have smooth light-gray bark; pole trees develop long shallow furrows; mature and old... Dormant: Opposite brown twigs carry a narrow, sharply pointed terminal bud with many tightly overlapping...	Tolerance: Very shade tolerant; seedlings and saplings can persist for years as advance... Site: Best on cool, fertile, moist but well-drained loams with adequate calcium and magnesium; performs...
Red maple <i>Acer rubrum</i>	Foliage: Opposite, simple blades usually have three to five lobes, angular or V-shaped sinuses, and regular serration... Bark: Young stems are smooth and light gray; intermediate bark develops narrow vertical cracks and small scaly plates;... Dormant: Opposite reddish twigs bear blunt-ovoid red buds with several pairs of scales; flower buds are...	Tolerance: Moderately shade tolerant when young and able to persist beneath partial canopy... Site: Occupies an exceptionally broad range from saturated swamps and floodplains to dry ridges; local seed...
American beech <i>Fagus grandifolia</i>	Foliage: Alternate, simple leaves are elliptic to ovate with a short petiole, straight parallel lateral veins, and one... Bark: Saplings and healthy mature trees have thin, smooth, pale bluish-gray bark, often compared with elephant hide... Dormant: Alternate twigs carry exceptionally long, slender, sharply pointed cigar-shaped buds with many...	Tolerance: Exceptionally shade tolerant; seedlings, saplings, and root suckers can persist... Site: Most competitive on moist, well-drained, often fertile loams and slopes; absent from persistently...
Yellow birch <i>Betula alleghaniensis</i>	Foliage: Alternate ovate leaves have a rounded to weakly heart-shaped base, many straight lateral veins, and sharply... Bark: Seedlings are gray-brown and unremarkable; intermediate stems develop bronze to yellow-gold bark with horizontal... Dormant: Slender alternate twigs bear pointed reddish-brown buds, small spur shoots, and no true terminal...	Tolerance: Intermediate in shade tolerance: more tolerant than paper birch and aspen, but... Site: Best on cool, moist, well-drained, fertile loams and coves; common on north and east slopes, seepage...
Paper birch <i>Betula papyrifera</i>	Foliage: Alternate ovate to triangular leaves have a rounded or wedge-shaped base, pointed tip, and doubly serrate... Bark: Young stems are reddish-brown with pale horizontal lenticels and do not look white; pole and mature bark becomes... Dormant: Slender reddish-brown alternate twigs have pointed appressed buds, conspicuous resin glands or...	Tolerance: Very shade intolerant; seedlings require abundant light and are rapidly overtopped... Site: Occupies cool sites across a broad moisture range, from well-drained glacial tills and sandy soils to...
Eastern hemlock <i>Tsuga canadensis</i>	Foliage: Short, flat, mostly blunt single needles vary in length along delicate twigs, show two pale stomatal bands... Bark: Young bark is smooth gray-brown; intermediate stems develop narrow scaly ridges; mature bark becomes dark gray... Dormant: Fine flexible yellow-brown twigs carry minute rounded buds, tiny petiolate needles, and often a...	Tolerance: Extremely shade tolerant, among the most tolerant North American trees; seedlings... Site: Favors cool, humid, sheltered sites with reliable moisture and generally well-drained acidic to...
American basswood <i>Tilia americana</i>	Foliage: Alternate, broadly ovate to heart-shaped leaves have an abruptly pointed tip, serrate margin, palmate basal... Bark: Young bark is smooth gray to gray-brown; intermediate bark develops narrow vertical fissures; old boles are dark... Dormant: Alternate zigzag twigs carry plump red to reddish-brown ovoid buds with only two or three...	Tolerance: Shade tolerant, though generally less persistent in deep shade than beech or sugar... Site: Best on deep, fertile, moist, well-drained loams with high base status, especially calcium-rich tills...
Black cherry <i>Prunus serotina</i>	Foliage: Alternate, simple, elliptic leaves have fine rounded or incurved teeth, one or more small glands near the top of... Bark: Young stems are smooth reddish-brown to dark gray with prominent horizontal lenticels; intermediate bark breaks... Dormant: Slender alternate reddish-brown twigs carry small ovoid scaled buds and semicircular leaf scars;...	Tolerance: Shade intolerant; seedlings may survive briefly under partial canopy, but... Site: Best on deep, fertile, moist but well-drained loams, especially Allegheny uplands and protected...
Eastern hophornbeam <i>Ostrya virginiana</i>	Foliage: Alternate ovate leaves have many straight lateral veins, a sharply pointed tip, and a doubly serrate margin; the... Bark: Young stems are smooth gray-brown with lenticels; older small trunks break into narrow, loose, vertical gray... Dormant: Slender alternate twigs bear short pointed ovoid buds with many visible scales; several...	Tolerance: Shade tolerant and able to persist as a slow-growing understory tree beneath... Site: Common on dry-mesic to mesic, well-drained, often rocky or calcareous soils; avoids prolonged flooding...
Black maple <i>Acer nigrum</i>	Foliage: Opposite blades are usually three-lobed, broad and somewhat drooping, with a deeply heart-shaped base, few... Bark: Young bark is gray and fairly smooth; older boles become dark gray-brown with irregular furrows and firm plates... Dormant: Opposite brown twigs carry a pointed terminal bud and smaller paired laterals. Bud, twig, and...	Tolerance: Very shade tolerant when young, broadly like sugar maple; suppressed advance... Site: Best on fertile, moist, well-drained loams and alluvium, often over calcareous or otherwise base-rich...
Striped maple <i>Acer pensylvanicum</i>	Foliage: Opposite, large, thin blades have three shallow forward-pointing lobes, a broad heart-shaped base, and fine... Bark: Saplings have smooth green bark marked by conspicuous vertical white stripes; older stems turn greenish brown to... Dormant: Opposite green to green-brown twigs and young stems retain longitudinal pale stripes; paired...	Tolerance: Very shade tolerant; survives decades suppressed but grows best under moderate... Site: Favors cool, moist, well-drained sandy loams, commonly acidic, and avoids both stagnant wetness and...
Mountain maple <i>Acer spicatum</i>	Foliage: Opposite blades are usually three-lobed, sometimes weakly five-lobed, with sharply pointed lobes and coarse... Bark: Young stems are reddish to gray-brown without white longitudinal stripes; older small trunks remain thin-barked... Dormant: Slender opposite twigs carry small pointed paired buds and a terminal bud; stems lack striped...	Tolerance: Moderately shade tolerant and able to persist beneath canopy, but density and... Site: Favors cool, moist, well-drained upland soils, rocky slopes, ravines, and streamside benches; drought...
Norway maple <i>Acer platanoides</i>	Foliage: Opposite, broad blades usually have five sharply pointed lobes and sparse large teeth. A freshly broken petiole... Bark: Young bark is smooth gray; mature bark develops regular narrow vertical ridges and shallow interlacing furrows... Dormant: Opposite stout twigs bear large plump terminal buds and paired lateral buds, commonly reddish...	Tolerance: Seedlings are shade tolerant to very shade tolerant and form persistent seedling... Site: Broadly site tolerant, performing especially well on cool moist rich soils while also tolerating urban...
Heart-leaved paper birch <i>Betula cordifolia</i>	Foliage: Alternate ovate leaves have a distinctly heart-shaped base, a pointed but not long-drawn tip, double teeth, and... Bark: Young stems are reddish to brown; mature bark becomes pink-white, brown-white, or red-brown-white and exfoliates... Dormant: Alternate twigs lack a true terminal bud and carry small ovoid imbricate buds; short shoots...	Tolerance: Generally shade intolerant to intermediate; seedlings and crowns perform best in... Site: Favors cool, moist, well-drained, often rocky or talus-derived upland soils and is poorly suited to...
Gray birch <i>Betula populifolia</i>	Foliage: Alternate glossy blades are triangular to rhombic-triangular with a long narrow tip, coarse double teeth, and... Bark: Young bark is brown to reddish; saplings and small trees become chalky gray-white with dark triangular chevrons... Dormant: Slender alternate twigs lack a true terminal bud, show lenticels and small resin glands, and...	Tolerance: Very shade intolerant; recruitment and persistence require open conditions. Site: Grows from moist well-drained margins to dry infertile sands, gravels, rocky slopes, and acidic...

Taxon	High-value visible evidence	Silvics anchor
Sweet birch <i>Betula lenta</i>	Foliage: Alternate ovate to narrow-ovate leaves are finely and regularly double-serrate, usually with at least six teeth... Bark: Young bark is smooth reddish brown with horizontal lenticels and can resemble cherry; mature bark becomes dark... Dormant: Slender reddish-brown twigs lack a true terminal bud and carry small pointed imbricate buds; a...	Tolerance: Classed as shade intolerant overall, although seedlings benefit from side shade or... Site: Best on moist well-drained soils and protected north- or east-facing slopes; tolerates rocky shallow...
Pin cherry <i>Prunus pensylvanica</i>	Foliage: Alternate thin leaves are narrow-ovate to lanceolate, long-pointed, finely serrate, and largely glabrous; one or... Bark: Young bark is shiny reddish brown with strong horizontal lenticels; older thin bark becomes gray-brown and may... Dormant: Slender reddish-brown twigs show horizontal lenticels and small sharp imbricate buds, including a...	Tolerance: Very shade intolerant; establishment and survival are concentrated in large openings. Site: Occupies coarse to medium soils from rocky ledges and sands to moist loams, but is generally absent...
Chokecherry <i>Prunus virginiana</i>	Foliage: Alternate oblong to obovate leaves have sharply pointed teeth, mostly 8-11 pairs of lateral veins, and a mostly... Bark: Young stems are reddish to gray-brown with horizontal lenticels; older small trunks remain relatively thin and... Dormant: Slender brown to reddish twigs have obvious lenticels and sharp imbricate buds. The winter twig...	Tolerance: Shade tolerant enough to persist under forest canopy, but reaches greatest density... Site: Broadly adaptable from mesic forest edges and riparian terraces to dry-mesic thickets, provided...
American hornbeam <i>Carpinus caroliniana</i>	Foliage: Alternate elliptic to ovate leaves have a symmetrical rounded or shallowly cordate base, strong straight... Bark: Young and mature bark remains thin, smooth, blue-gray to slate gray, while the trunk develops distinctive sinewy... Dormant: Slender alternate twigs lack a true terminal bud; narrow ovoid buds have many overlapping scales...	Tolerance: Very shade tolerant as a seedling and sapling; tolerance declines with age, so... Site: Best on rich loams with high water-holding capacity, from somewhat poorly drained to well drained...
Pagoda dogwood <i>Cornus alternifolia</i>	Foliage: Leaves are alternate, though crowded near shoot tips, with entire to slightly wavy margins and several strong... Bark: Young stems are smooth reddish to green-brown; mature bark becomes gray-brown with shallow ridges or small... Dormant: Leaves and buds are alternate, often clustered near upturned branch tips; slender twigs are...	Tolerance: Shade tolerant and able to dominate locally in mature-forest understories, while... Site: Favors cool, rich, moist, well-drained forest soils and edges; drought and heat stress increase decline.
Smooth serviceberry <i>Amelanchier laevis</i>	Foliage: Alternate ovate to elliptic leaves are sharply toothed; at flowering they are already partly expanded and... Bark: Young stems and twigs are reddish; older bark remains thin and smooth gray with pale vertical stripes, later... Dormant: Slender alternate reddish-brown twigs have narrow pointed imbricate buds, a terminal bud,...	Tolerance: Tolerates understory shade but flowers, fruits, and develops best with partial sun... Site: Prefers moist, moderately to well-drained acidic to neutral soils and tolerates rocky forest sites if...
Balsam fir <i>Abies balsamea</i>	Foliage: Needles are single, soft, flattened, aromatic when crushed, and usually arranged in two lateral ranks on lower... Bark: Sapling bark is thin, smooth, gray, and commonly dotted with conspicuous raised resin blisters. Intermediate... Dormant: Slender gray-brown twigs are relatively smooth after needles fall, with round flush leaf scars...	Tolerance: Very shade tolerant; seedlings can persist for decades as advance regeneration and... Site: Occupies moist cool sites from poorly drained organic and fine-textured lowlands to well-drained...
White spruce <i>Picea glauca</i>	Foliage: Needles are single, stiff, sharp, four-angled to somewhat flattened, and roll between the fingers; each stands... Bark: Young bark is thin, smoothish, gray to brown, becoming scaly early. Intermediate and mature bark forms thin... Dormant: Current branchlets are light brown to pink-brown, usually hairless, and coated with a thin...	Tolerance: Intermediate in shade tolerance; more tolerant than jack or red pine and the... Site: Grows across a wide moisture range but performs best on moist, well-drained, relatively fertile loams...
Black spruce <i>Picea mariana</i>	Foliage: Short single needles are four-angled, stiff, dull gray- to blue-green, glaucous, and borne around the twig on... Bark: Saplings have thin gray-brown bark with fine scales; intermediate and mature stems become dark gray to reddish... Dormant: Young yellow-brown branchlets lack bloom and are minutely hairy; at least some hairs are gland...	Tolerance: Shade tolerant, especially as a seedling, but generally less tolerant than balsam... Site: Dominant on cold acidic peatlands, poorly drained organic soils, and nutrient-poor mineral sites,...
Red spruce <i>Picea rubens</i>	Foliage: Single sharp four-angled needles stand around the twig on woody pegs and roll between the fingers. They are... Bark: Young bark is smoothish gray-brown to reddish brown and soon develops thin flaky scales. Mature bark becomes... Dormant: Yellow-brown current branchlets are minutely hairy without a glaucous bloom; the hairs are...	Tolerance: Very shade tolerant, broadly comparable to balsam fir and more tolerant than white... Site: Best in cool humid climates on moist acidic, commonly shallow or stony soils with thick organic...
Norway spruce <i>Picea abies</i>	Foliage: Needles are single, stiff, sharp, four-angled, dark green, and borne on woody pegs around the shoot. They... Bark: Young bark is smoothish gray-brown to reddish brown, becoming finely scaly. Older trunks develop gray to brown... Dormant: Reddish-brown branchlets carry prominent woody needle pegs and pointed reddish-brown, usually...	Tolerance: Moderately shade tolerant as a young tree, but plantation growth and crown... Site: Prefers cool, moist, well-drained, commonly acidic loams or sandy loams; prolonged saturation, heat,...
Eastern white pine <i>Pinus strobus</i>	Foliage: Soft, slender, flexible blue-green needles occur in fascicles of five, one for each letter in 'white'; they are... Bark: Seedlings and young saplings have smooth thin gray-green bark. Intermediate stems become gray with shallow... Dormant: Twigs are slender and gray-brown, with a conspicuous terminal cluster of long narrow pointed...	Tolerance: Intermediate in shade tolerance; seedlings tolerate substantial shelter, but... Site: Occurs on many soils and is competitive on sandy, rocky, or glacial deposits; best growth is on moist,...
Red pine <i>Pinus resinosa</i>	Foliage: Needles occur in fascicles of two, are dark green, relatively straight, and usually 9-16 cm long. They are... Bark: Young stems have thin gray to reddish scaly bark. Intermediate and mature boles develop conspicuous orange-red... Dormant: Stout orange-brown branchlets end in clustered ovoid red-brown buds, usually resinous. Persistent...	Tolerance: Shade intolerant; seedlings and saplings need nearly full light, and crowns recede... Site: Best known on well-drained sands, gravels, and rocky ridges, though highest productivity occurs on...
Jack pine <i>Pinus banksiana</i>	Foliage: Needles occur in fascicles of two, are short, stiff, yellow- to gray-green, and usually 2-5 cm long. Each pair... Bark: Sapling bark is thin, dull gray to reddish brown, and finely scaly. Mature stems remain relatively thin-barked... Dormant: Slender gray-brown twigs bear small ovoid resinous buds and persistent paired fascicle sheaths...	Tolerance: Very shade intolerant, among the least tolerant northern conifers; seedlings... Site: Most competitive on dry nutrient-poor sands, dunes, outwash, and barrens, but also occurs on rocky...
Scots pine <i>Pinus sylvestris</i>	Foliage: Needles occur in fascicles of two, are stiff, noticeably twisted, often blue- or gray-green, and usually 3-7 cm... Bark: Young branches and upper trunks develop thin flaky cinnamon- to bright orange-brown bark, a major field clue... Dormant: Gray- to orange-brown twigs bear clustered ovoid pointed gray-brown to reddish buds that may...	Tolerance: Shade intolerant; regeneration and good crown growth require open sun. Site: Adaptable to nutrient-poor, acidic, sandy, and seasonally dry soils if drainage is good; it performs...
Tamarack <i>Larix laricina</i>	Foliage: Soft, slender, blue- to bright-green needles occur singly on elongating current shoots and in dense tufts of... Bark: Young bark is smoothish gray to reddish brown and soon becomes finely scaly. Mature bark sheds as small thin... Dormant: Winter crowns are completely leafless but retain numerous rounded buds on short dark knobby spur...	Tolerance: Very shade intolerant; seedlings may endure light shade briefly but must reach the... Site: Characteristic of bogs, fens, and cold swamps on organic soils, yet best growth often occurs on moist,...
European larch <i>Larix decidua</i>	Foliage: Soft bright-green deciduous needles occur singly on long shoots and in dense tufts on knobby short shoots... Bark: Young branch bark is smooth gray to yellow-brown. Mature trunks develop thick reddish- to gray-brown, deeply... Dormant: Leafless winter twigs carry many rounded buds on stout knobby spur shoots; lateral branchlets...	Tolerance: Very shade intolerant; best stem and crown development require full sun, though... Site: Prefers cool sites with moist, well-drained, moderately fertile loams or gravelly soils; waterlogging,...

Taxon	High-value visible evidence	Silvics anchor
Northern white-cedar <i>Thuja occidentalis</i>	Foliage: Tiny opposite scale leaves overlap in four ranks and completely hide the twig, forming strongly flattened fan-... Bark: Sapling bark is thin, smoothish, reddish to gray-brown, and begins peeling in narrow fibers. Mature and old bark... Dormant: Evergreen flattened sprays persist, with no conspicuous scaled winter bud; growth ends in tightly...	Tolerance: Very shade tolerant, able to persist beneath mixed canopies, although successful... Site: Occupies a broad moisture range from swamps and fens to dry cliffs, commonly favored by calcareous or...
Eastern redcedar <i>Juniperus virginiana</i>	Foliage: Adult branchlets carry tiny opposite scale leaves tightly appressed in four ranks, forming rounded to four-... Bark: Young bark is thin reddish brown to gray and begins peeling early. Mature bark shreds in long narrow fibrous... Dormant: Evergreen scale leaves or prickly juvenile leaves persist, and there is no large conspicuous...	Tolerance: Shade intolerant to only weakly tolerant; seedlings establish in open ground and... Site: Extremely site-flexible on dry rocky, sandy, clayey, and calcareous soils, including shallow...
Quaking aspen <i>Populus tremuloides</i>	Foliage: Alternate, simple leaves on normal short shoots are usually nearly round to broadly ovate or rhombic, 2-8 cm... Bark: Young stems have thin, smooth, pale greenish-white to cream bark, often with dark branch scars and horizontal... Dormant: Alternate slender brown to gray-red twigs end in narrow, pointed, usually glabrous buds with...	Tolerance: Very shade intolerant; seedlings and root suckers require nearly full light, and... Site: Occupies a very broad edaphic range, from dry sands and shallow uplands to moist loams and cold...
Bigtooth aspen <i>Populus grandidentata</i>	Foliage: Alternate, simple blades on normal shoots are broadly ovate, usually 7-12 cm long, and bear roughly 5-12... Bark: Saplings have smooth thin olive-green to gray bark with lenticels and dark branch scars. Pole bark becomes dull... Dormant: Alternate gray-brown twigs have a true terminal bud and appressed lateral buds. The several...	Tolerance: Very shade intolerant; successful suckers and seedlings need nearly full overhead... Site: Most characteristic on well-drained sandy or loamy uplands and glacial deposits, but best growth...
Balsam poplar <i>Populus balsamifera</i>	Foliage: Alternate, simple blades are broadly ovate to lance-ovate with a long acute or short-acuminate tip, a broad-... Bark: Young stems have smooth gray-green to brown bark with evident lenticels; pole bark remains relatively smooth and... Dormant: The large, sharply pointed terminal bud is conspicuously resinous, sticky, and strongly balsamic...	Tolerance: Very shade intolerant; regeneration needs nearly full light and is rapidly... Site: Best developed on moist, nutrient-rich, aerated alluvial soils and fresh flood deposits; it also...
Speckled alder <i>Alnus incana subsp. rugosa</i>	Foliage: Alternate, simple leaves are narrow-ovate to elliptic or broadly elliptic, commonly with a rounded to subcordate... Bark: Young brown to gray stems are smooth and densely speckled with conspicuous horizontal or oval white lenticels,... Dormant: Alternate ovoid buds stand on obvious 2-4 mm stalks and are covered by two, sometimes three...	Tolerance: Generally shade intolerant to intermediate; dense thickets persist where crowns... Site: Characteristic of saturated or seasonally flooded mineral and organic soils along streams, shorelines...
American mountain-ash <i>Sorbus americana</i>	Foliage: Leaves are alternate and pinnately compound, generally 10-22 cm long, with a terminal leaflet and numerous... Bark: Young trunks are thin, smooth, gray to gray-brown, with horizontal lenticels; older small boles become somewhat... Dormant: Alternate twigs carry a conspicuous ovoid to pointed terminal bud and smaller lateral buds with...	Tolerance: Shade intolerant overall; it can persist beneath a semiopen canopy but abundance... Site: Most successful on cool moist acidic soils with adequate aeration, but occurs from swamp borders to...
Showy mountain-ash <i>Sorbus decora</i>	Foliage: Alternate pinnately compound leaves commonly carry 13-17 blue-green leaflets. Compared with American mountain-... Bark: Young gray-brown bark is smooth with lenticels; older small boles become scaly or shallowly fissured but remain... Dormant: Alternate twigs end in a conspicuous ovoid to pointed scaly bud. The margins of the winter bud...	Tolerance: Tolerates partial shade but not a deeply closed canopy; crown growth, flowering... Site: Favored by cool moist conditions and occurs on a variety of acidic to neutral substrates, including...
Silver maple <i>Acer saccharinum</i>	Foliage: Leaves are opposite, simple, and usually five-lobed, with very deep sinuses that produce long narrow lobes; the... Bark: Seedlings and saplings have smooth light-gray bark, sometimes with fine vertical lenticels. Intermediate stems... Dormant: Twigs are opposite, slender to moderately stout, reddish brown, and bear blunt red to reddish...	Tolerance: Intermediate to moderately tolerant when young, but best crown development and... Site: Most characteristic on moist alluvial soils and seasonally flooded lowlands; it tolerates considerable...
Boxelder <i>Acer negundo</i>	Foliage: Leaves are opposite and pinnately compound, an unusual combination among regional trees. Each leaf usually has... Bark: Young bark is smooth, thin, and gray to light brown. Intermediate stems develop shallow narrow fissures and low... Dormant: Opposite stout twigs are green, olive, purplish, or brown and often carry a pale waxy bloom. Buds...	Tolerance: Shade intolerant to intermediate when young; it persists best along edges and in... Site: Most frequent on moist fertile alluvium, stream margins, and low terraces, but remarkably tolerant of...
River birch <i>Betula nigra</i>	Foliage: Leaves are alternate, simple, and broadly rhombic to ovate, usually with a wedge-shaped base, acute tip, and... Bark: Young stems and upper branches have salmon, cinnamon, pinkish tan, or cream bark peeling in thin papery curls... Dormant: Slender alternate twigs are reddish brown to gray, lenticellate, and may bear glandular hairs...	Tolerance: Shade intolerant; seedlings need open light and established crowns deteriorate... Site: Best on moist, well-aerated alluvial sands, silts, and loams along waterways; it tolerates periodic...
White ash <i>Fraxinus americana</i>	Foliage: Leaves are opposite and pinnately compound, usually with seven but sometimes five to nine stalked leaflets... Bark: Seedlings and young saplings have smooth gray bark with scattered pale lenticels. Pole-size bark develops... Dormant: Stout opposite brown to blue-brown twigs are normally glabrous. The terminal bud is rounded at...	Tolerance: Young seedlings tolerate shade for a time, but tolerance declines with age... Site: Best on deep, fertile, moist but well-drained loams with favorable nutrient status; it avoids...
Green ash <i>Fraxinus pennsylvanica</i>	Foliage: Leaves are opposite and pinnately compound with five to nine stalked leaflets. Leaflets are usually serrate and... Bark: Young bark is smooth gray-brown with lenticels. Pole and mature stems develop narrow interlacing ridges and... Dormant: Opposite gray-brown branchlets are usually pubescent on current growth, and the terminal bud is...	Tolerance: Variable from intermediate to moderately tolerant when young, with advance... Site: Most characteristic on fertile alluvial clays, silts, and loams subject to periodic flooding, yet...
Black ash <i>Fraxinus nigra</i>	Foliage: Leaves are opposite and pinnately compound with seven to thirteen, occasionally more, narrow sharply serrate... Bark: Sapling bark is smooth light gray and may feel somewhat corky. Intermediate stems develop shallow fissures and... Dormant: Stout opposite twigs bear dark brown to nearly black buds. The terminal bud stands distinctly...	Tolerance: Intermediate when young, with seedlings able to persist under a partial canopy but... Site: Specialized for cold, poorly drained mineral or organic soils, often where nutrient-rich groundwater...
Eastern cottonwood <i>Populus deltoides</i>	Foliage: Leaves are alternate, simple, broadly triangular or deltoid, and often nearly as wide as long, with a truncate... Bark: Seedlings and young branches are smooth yellow-green to gray with lenticels. Intermediate trunks become gray and... Dormant: Twigs are stout, glossy yellow-brown to reddish brown, conspicuously lenticellate, and often...	Tolerance: Very shade intolerant; seedlings and lower crowns decline rapidly when overtopped. Site: Best on deep moist, well-aerated sands and silts deposited along rivers; tolerates periodic flooding...
American elm <i>Ulmus americana</i>	Foliage: Leaves are alternate, simple, usually seven to fourteen centimeters long, ovate to elliptic, abruptly pointed... Bark: Young bark is smooth gray to gray-brown. Intermediate stems form shallow intersecting fissures and flattened... Dormant: Slender alternate twigs form a slight zigzag and generally lack corky wings. Small ovoid brown...	Tolerance: Intermediate to moderately tolerant when young; seedlings can persist beneath a... Site: Broad site amplitude from moist well-drained uplands to seasonally flooded rich alluvium, with best...
Slippery elm <i>Ulmus rubra</i>	Foliage: Leaves are alternate, simple, usually eight to sixteen centimeters long, broad ovate to elliptic, double-... Bark: Young bark is gray-brown and shallowly fissured. Intermediate and mature bark becomes dark gray to reddish brown... Dormant: Alternate zigzag twigs are more often stout and hairy than those of American elm. Ovoid buds have...	Tolerance: Intermediate to tolerant when young, allowing seedlings to persist beneath partial... Site: Best on moist fertile, well-drained loams and calcareous rich woods; occurs on riparian terraces and...

Taxon	High-value visible evidence	Silvics anchor
Rock elm <i>Ulmus thomasi</i>	Foliage: Leaves are alternate, simple, elliptic to obovate or ovate, double-serrate, and usually firm textured, with a... Bark: Young stems are smooth gray-brown. Intermediate bark develops shallow fissures and narrow firm ridges. Mature... Dormant: Alternate twigs are gray, brown, or reddish and lack a true terminal bud. Two- to several-year...	Tolerance: Intermediate in shade tolerance; seedlings persist under partial canopy but need... Site: Most characteristic on well-drained loams, rocky uplands, and calcareous substrates, though it also...
Siberian elm <i>Ulmus pumila</i>	Foliage: Leaves are alternate, simple, small, usually two to six and a half centimeters long, elliptic to lanceolate, and... Bark: Young bark is smooth gray to light brown. Intermediate stems develop shallow fissures and narrow ridges. Mature... Dormant: Slender alternate gray-brown twigs form a zigzag and lack a true terminal bud. Small globose buds...	Tolerance: Shade intolerant; seedlings establish and grow best in open fields, roadsides,... Site: Extremely tolerant of drought, cold, wind, alkaline or nutrient-poor soils, and compaction, while also...
Black willow <i>Salix nigra</i>	Foliage: Leaves are alternate, simple, narrowly lanceolate, mostly about 7.5-17 millimeters wide, long-pointed, and... Bark: Young bark is smooth yellow-brown, gray, or reddish brown. Intermediate stems become shallowly fissured with... Dormant: Slender alternate branchlets are yellow, orange, red-brown, or gray, conspicuously lenticellate,...	Tolerance: Very shade intolerant; seedlings and lower crowns decline rapidly beneath taller... Site: Requires a continuous growing-season water supply and is most competitive on very wet alluvial silts,...
American sycamore <i>Platanus occidentalis</i>	Foliage: Leaves are alternate, simple, broad, palmately veined, and usually three to five shallow or moderate lobes with... Bark: Young stems are gray-brown and smoothish. Intermediate trunks begin exfoliating in irregular thin plates that... Dormant: Stout alternate twigs zigzag between nodes and show a complete stipule-scar ring encircling each...	Tolerance: Very shade intolerant; seedlings need open light and crowns self-prune or decline... Site: Best on deep moist, fertile, well-aerated alluvial loams and sands along streams; tolerates periodic...

Part III - Fillable instructional sessions

Complete sessions in order when possible. Closed-book retrieval means close or hide the teaching atlas before answering.

Session A - Regional red-oak group

Purpose: character-combination teaching across sun and shade foliage. Taxon scope: listed new taxa. Feedback: during instruction.

Before the session: which species or comparison currently feels least reliable?

Northern red oak: one diagnostic character plus one silvics anchor

Black oak: one diagnostic character plus one silvics anchor

Scarlet oak: one diagnostic character plus one silvics anchor

Northern pin oak: one diagnostic character plus one silvics anchor

Pin oak: one diagnostic character plus one silvics anchor

White oak: one diagnostic character plus one silvics anchor

Swamp white oak: one diagnostic character plus one silvics anchor

Bur oak: one diagnostic character plus one silvics anchor

Chestnut oak: one diagnostic character plus one silvics anchor

Common hackberry: one diagnostic character plus one silvics anchor

Exit retrieval: state the most important correction or strengthened distinction

☐ Session A complete (enter X)

Session B - Red-oak retrieval

Purpose: unfamiliar isolated organs and whole-tree decisions. Taxon scope: session a taxa plus midterm remediation. Feedback: after batch submission.

Before the session: which species or comparison currently feels least reliable?

1. Silvics

Design the species-specific part of a prescription for White ash: state what to protect or control, and justify the cutting or regeneration approach from its known response.

Response

closed-book retrieval

2. Silvics

For Pin oak (*Quercus palustris*), describe the moisture, drainage, and soil setting a forester should expect, then place it in its climatic or forest-association context.

Response

closed-book retrieval

3. Silvics

A stand prescription lists Black cherry as an objective species. State the site conditions that favor it and one site or climatic constraint that would make that objective questionable.

Response

closed-book retrieval

4. Silvics

A stand prescription lists Mountain maple as an objective species. State the site conditions that favor it and one site or climatic constraint that would make that objective questionable.

Response

closed-book retrieval

5. Silvics

Predict how Showy mountain-ash responds to disturbance or release, and identify one browse, insect, disease, fire, flooding, or injury concern a field forester should carry into the stand.

Response

closed-book retrieval

6. Silvics

A stand containing Swamp white oak is disturbed. Explain the likely biological response and one important damaging agent or vulnerability that changes the prognosis.

Response

closed-book retrieval

7. Forest Association

A field crew reports Balsam poplar from an unfamiliar stand. State the regional and community context that would make the report credible, then request morphological evidence needed to confirm it.

Response

closed-book retrieval

8. Silvics

A stand prescription lists Jack pine as an objective species. State the site conditions that favor it and one site or climatic constraint that would make that objective questionable.

Response

closed-book retrieval

9. Forest Association

A field crew reports Swamp white oak from an unfamiliar stand. State the regional and community context that would make the report credible, then request morphological evidence needed to confirm it.

Response

closed-book retrieval

10. Silvics

Explain how Red maple normally regenerates, including establishment requirements and any seed, mast, sprouting, layering, or suckering behavior that materially affects a prescription.

Response

closed-book retrieval

Exit retrieval: state the most important correction or strengthened distinction

☐ Session B complete (enter X)

Session C - White-oak group and hackberry

Purpose: direct group contrasts followed by mixed isolated specimens. Taxon scope: listed new taxa plus cs11 and cs12. Feedback: after response.

Before the session: which species or comparison currently feels least reliable?

Sugar / black / Norway maple

Course separator: Petiole latex; leaf underside hair and stipules; bud size/shape; samara angle

My visible separator

Best resolving view

Red / silver maple

Course separator: Sinus depth; lower-surface whiteness; spring samara size; twig odor; bark strips

My visible separator

Best resolving view

Striped / mountain maple

Course separator: Green-white striped bark versus unstriped; leaf size/base; fruit cluster posture; growth form

My visible separator

Best resolving view

White / green / black ash

Course separator: Leaf-scar shape and bud position; leaflet stalks; samara wing extent; black-ash corky bark

My visible separator

Best resolving view

Paper / heart-leaved / gray birch

Course separator: Leaf base and vein count; twig resin glands; peeling versus tight bark; black branch marks; elevation

My visible separator

Best resolving view

Yellow / sweet birch

Course separator: Both wintergreen; yellow birch golden curling bark and spur shoots versus sweet birch dark blocky bark

My visible separator

Best resolving view

White / black / red / Norway spruce

Course separator: Cone size and scale margin; twig hairs; bud resin; foliage color/odor; pendulous branchlets

My visible separator

Best resolving view

Spruce / fir / hemlock

Course separator: Woody pegs versus pad base versus petiole; needle cross-section; cone orientation and persistence

My visible separator

Best resolving view

Red / jack / Scots / pitch pine

Course separator: Fascicle count/length; bend test; cone curvature/prickles; upper-bole orange bark; epicormic shoots

My visible separator

Best resolving view

Tamarack / European larch

Course separator: Cone size and scale count/edge; needle cluster density; native peatland versus planting

My visible separator

Best resolving view

Red-oak group

Course separator: Bud hair/size; lower-leaf hair; sinus architecture; acorn cup depth/rings; bark; site and branch angle

My visible separator

Best resolving view

White-oak group

Course separator: Lower-leaf felt; fruit stalk; fringed cup; chestnut teeth; bark architecture; site

My visible separator

Best resolving view

Aspens / balsam poplar / cottonwood

Course separator: Petiole flattening; tooth size; leaf base; bud size/stickiness/odor; bark and floodplain/upland context

My visible separator

Best resolving view

American / slippery / rock / Siberian elm

Course separator: Leaf roughness/base; bud hair; corky wings; samara hair/notch; bark layering; crown

My visible separator

Best resolving view

Black / pin / chokecherry

Course separator: Fruit cluster architecture; petiole glands; lower-midvein hair; bark age; whole pioneer/clonal form

My visible separator

Best resolving view

American hornbeam / eastern hophornbeam

Course separator: Muscular smooth bole and three-lobed fruit bracts versus shreddy bark and hoplike sacs

My visible separator

Best resolving view

Yellow birch / black cherry bark

Course separator: Golden horizontal curls and wintergreen twig versus burnt-chip plates, petiole glands, and almond twig odor

My visible separator

Best resolving view

Serviceberry / mountain-ashes / chokecherry

Course separator: Simple versus compound leaves; pome calyx versus drupe; leaflet proportions; buds; fruit architecture

My visible separator

Best resolving view

Exit retrieval: state the most important correction or strengthened distinction

☐ Session C complete (enter X)

Session D - Oak multi-organ laboratory

Purpose: leaves buds acorns bark and site stations. Taxon scope: all module taxa. Feedback: after station batch submission.

Before the session: which species or comparison currently feels least reliable?

Practice 1

Identity: common OR scientific name



Evidence asset INAT_PHOTO_495802266 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 2

Identity: common OR scientific name



Evidence asset INAT_PHOTO_701470945 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 3

Identity: common OR scientific name



Evidence asset INAT_PHOTO_715677057 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 4

Identity: common OR scientific name



Evidence asset INAT_PHOTO_720938161 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 5

Identity: common OR scientific name



Evidence asset INAT_PHOTO_699497364 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 6

Identity: common OR scientific name



Evidence asset INAT_PHOTO_643224944 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

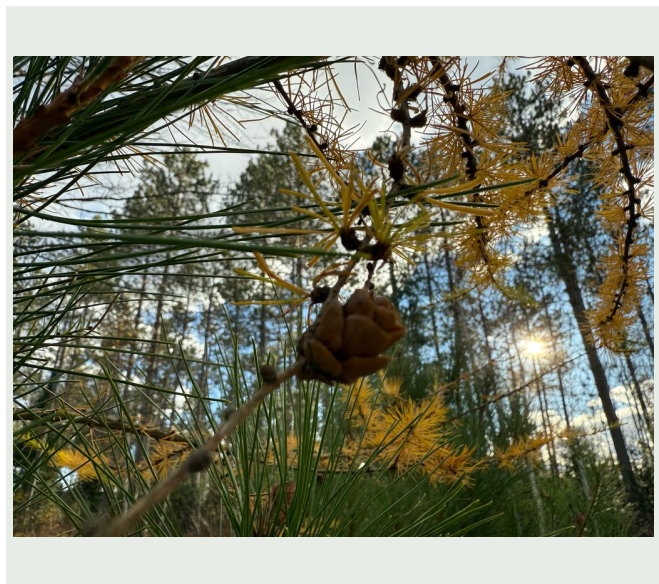
Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 7

Identity: common OR scientific name



Evidence asset INAT_PHOTO_587045770 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 8

Identity: common OR scientific name



Evidence asset INAT_PHOTO_699773690 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 9

Identity: common OR scientific name



Evidence asset INAT_PHOTO_719216159 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 10

Identity: common OR scientific name



Evidence asset INAT_PHOTO_598619493 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric calls for a field decision.
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Exit retrieval: state the most important correction or strengthened distinction

☐ Session D complete (enter X)

Session E - Oak-transition cumulative retrieval

Purpose: adaptive retrieval with nomenclature and silvics prompts. Taxon scope: cr 01 eligible taxa prioritized by mastery. Feedback: after batch submission.

Before the session: which species or comparison currently feels least reliable?

1. Forest Association

A field crew reports Bigtooth aspen from an unfamiliar stand. State the regional and community context that would make the report credible, then request morphological evidence needed to confirm it.

Response

closed-book retrieval

2. Forest Association

A field crew reports Bur oak from an unfamiliar stand. State the regional and community context that would make the report credible, then request morphological evidence needed to confirm it.

Response

closed-book retrieval

3. Family Relationship

To which botanical family does Smooth serviceberry (*Amelanchier laevis*) belong?

Response

closed-book retrieval

4. Silvics

A stand containing Sugar maple is disturbed. Explain the likely biological response and one important damaging agent or vulnerability that changes the prognosis.

Response

closed-book retrieval

5. Nomenclature

Translate this preferred common name into the canonical binomial: Mountain maple.

Response

closed-book retrieval

6. Forest Association

A field crew reports White spruce from an unfamiliar stand. State the regional and community context that would make the report credible, then request morphological evidence needed to confirm it.

Response

closed-book retrieval

7. Nomenclature

Give the preferred course common name for *Acer nigrum*.

Response

closed-book retrieval

8. Nomenclature

Give the accepted scientific name for Balsam poplar.

Response

closed-book retrieval

9. Silvics

Predict the role of Sweet birch from suppression through canopy recruitment. Tie the prediction to shade tolerance, successional status, and growth or longevity.

Response

closed-book retrieval

10. Forest Association

A field crew reports White ash from an unfamiliar stand. State the regional and community context that would make the report credible, then request morphological evidence needed to confirm it.

Response

closed-book retrieval

Exit retrieval: state the most important correction or strengthened distinction

☐ Session E complete (enter X)

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Species-profile references

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